



American  
Brain Tumor  
Association®

2015 NATIONAL PATIENT & FAMILY CONFERENCE

## Providing & Pursuing Answers: Advances in Research, Treatment and Care



**Ronald E. Warnick, MD**

*Medical Director*

*University of Cincinnati Brain Tumor Center*

## Meningiomas: Update on Treatment and Care

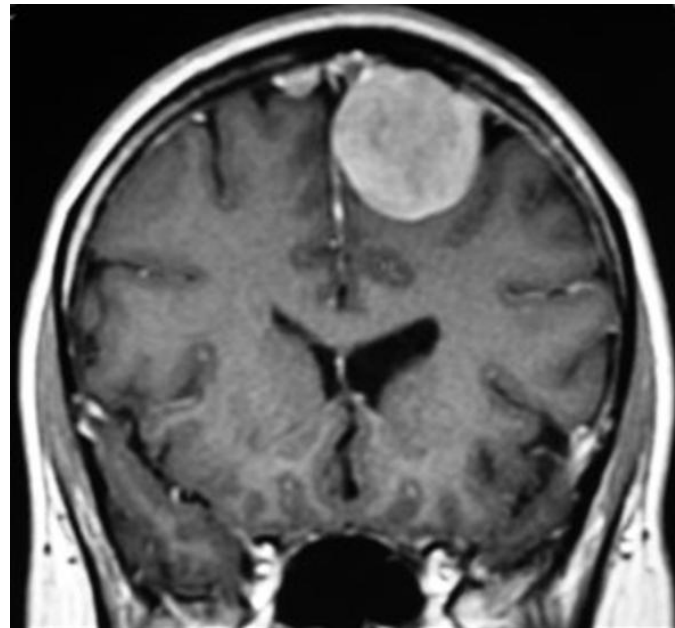
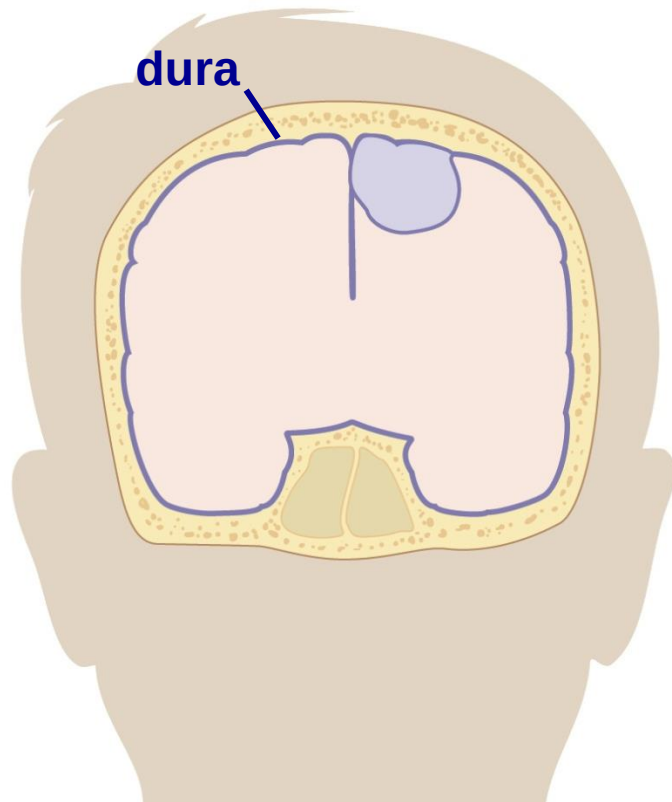


# Overview

- Incidence
- Risk Factors
- Classification
- Presenting Symptoms
- Imaging Modalities
- Surgery
- Radiotherapy
- Medical Therapy
- Future Advances

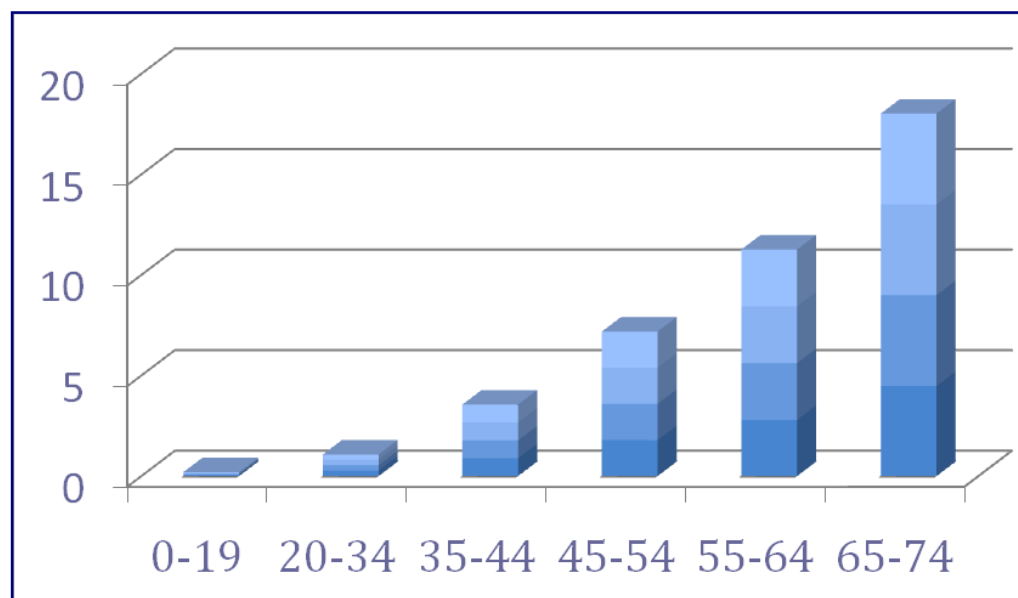
# Origin

**Meningiomas arise from the dura (the outer-most protective layer of the brain)**



# Incidence

- Meningiomas represent 36% of all primary brain tumors (25,610 per year)
- More common in women (3:1)
- Average age = 63 years



# Incidence

| Age     | Most Common Brain Tumor |       |
|---------|-------------------------|-------|
|         | Tumor Type              | Rate  |
| 0-4     | Embryonal Tumors        | 1.23  |
| 5-9     | Pilocytic Astrocytoma   | 0.96  |
| 10-14   | Pilocytic Astrocytoma   | 0.85  |
| 15-19   | Tumors of the Pituitary | 1.50  |
| 20-34   | Tumors of the Pituitary | 2.97  |
| 35-44   | Meningioma              | 4.66  |
| 45-54   | Meningioma              | 8.79  |
| 55-64   | Meningioma              | 14.35 |
| 65-74   | Meningioma              | 25.08 |
| 75-84   | Meningioma              | 37.49 |
| 85+     | Meningioma              | 49.48 |
| OVERALL | Meningioma              | 7.61  |

# Risk Factors

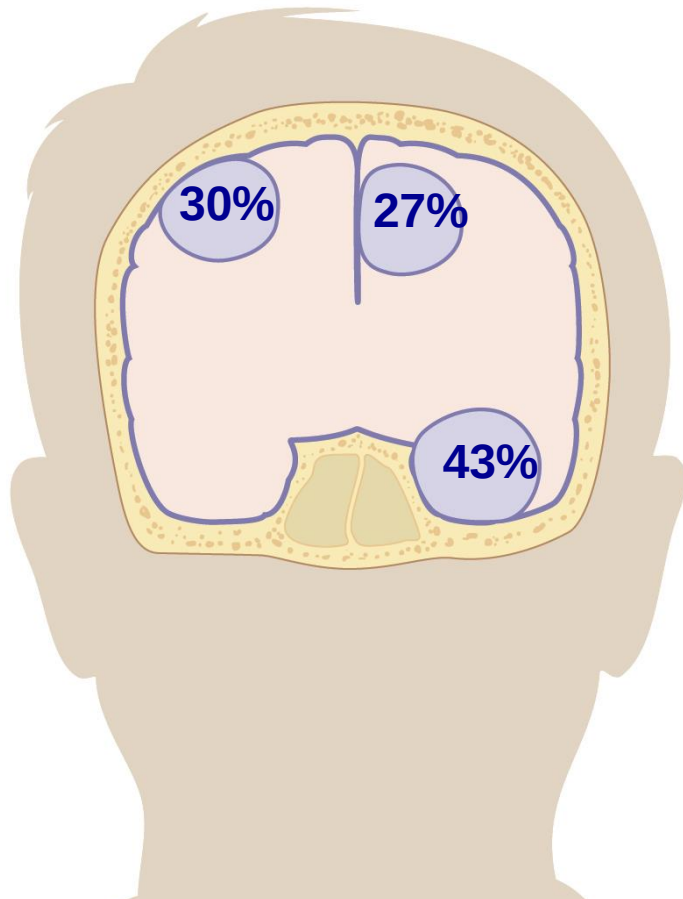
## ■ Definite

- Higher level radiation (Israeli experience)
- Genetic disorders (Neurofibromatosis 2)

## ■ Possible

- Lower level radiation (dental X-rays, cell phones)
- Head trauma (concussion)
- Breast cancer association
- Allergies (protective effect)

# Classification: *By Distribution*

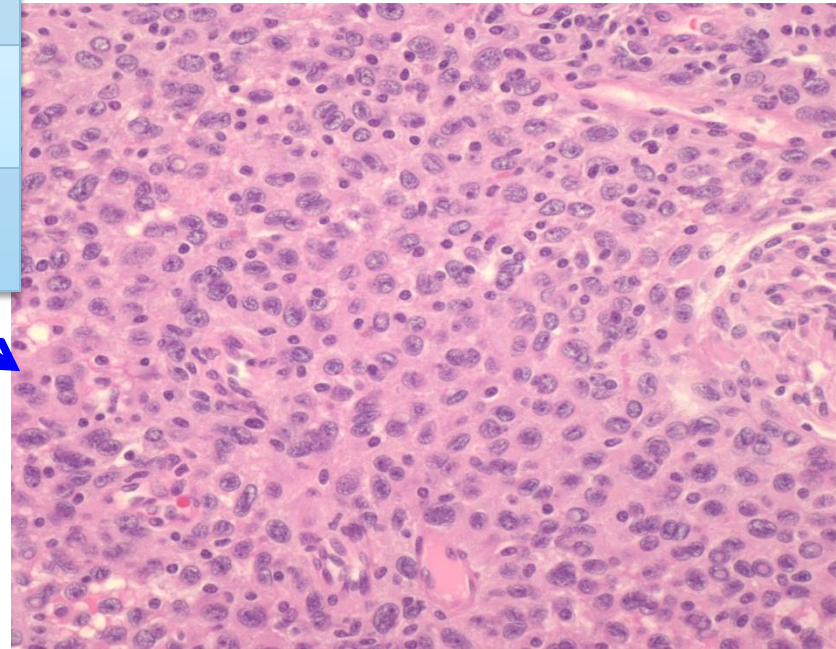
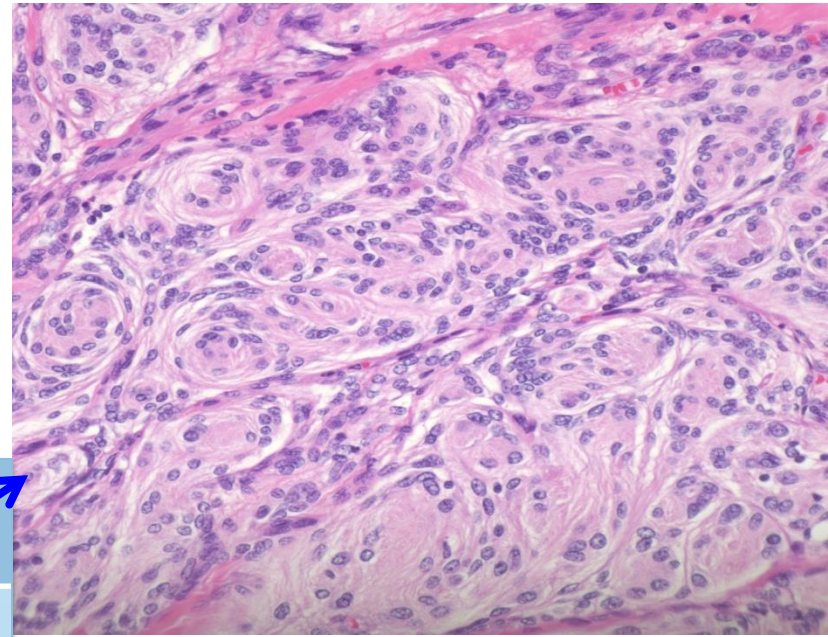


| Location   | Frequency |
|------------|-----------|
| Convexity  | 30%       |
| Midline    | 27%       |
| Skull Base | 43%       |

**Incidence of multiple meningiomas = 8%**

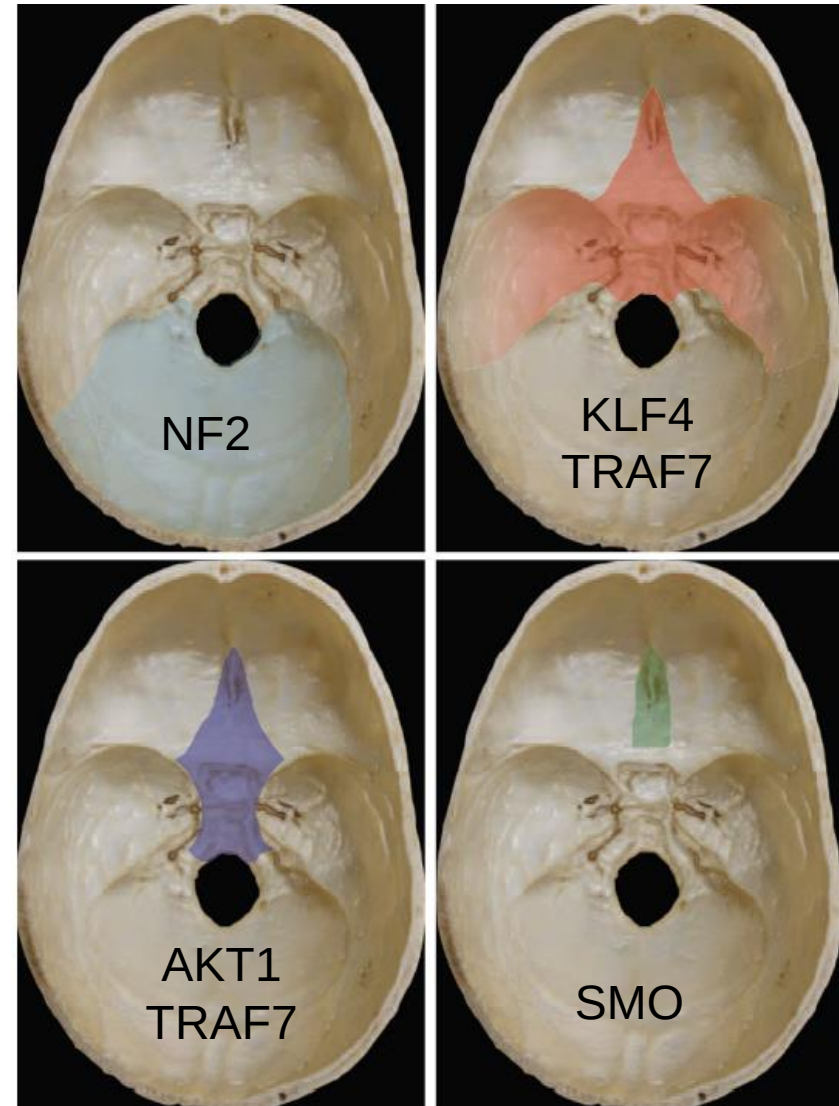
# Classification: *By Pathology*

| Grade | Description |
|-------|-------------|
| I     | Benign      |
| II    | Atypical    |
| III   | Malignant   |



# Classification: *By Genetics*

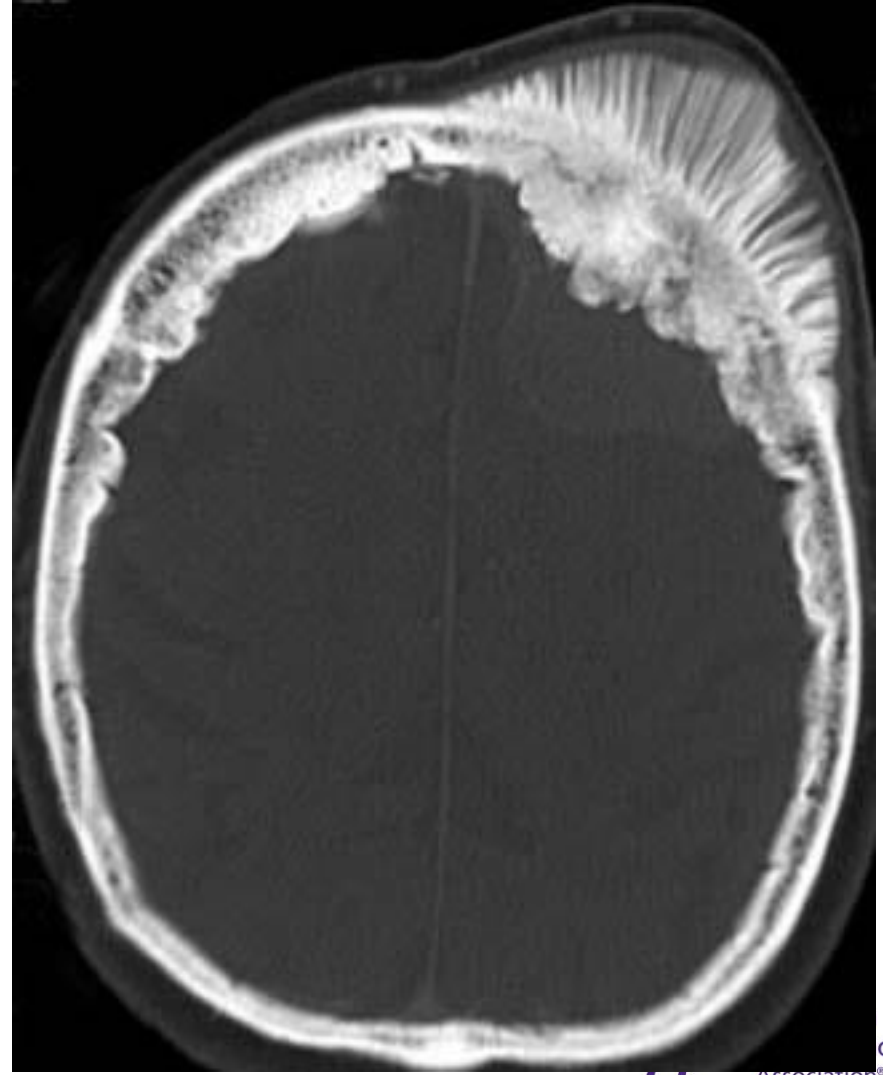
- **Loss of the NF2 gene is found in ~50% of sporadic meningiomas**



# Presenting Symptoms

| Symptom   | Frequency |
|-----------|-----------|
| Headache  | 70%       |
| Seizure   | 54%       |
| Cognitive | 52%       |
| Weakness  | 43%       |
| Speech    | 27%       |
| Sensation | 14%       |
| Vision    | 6%        |

# External Signs



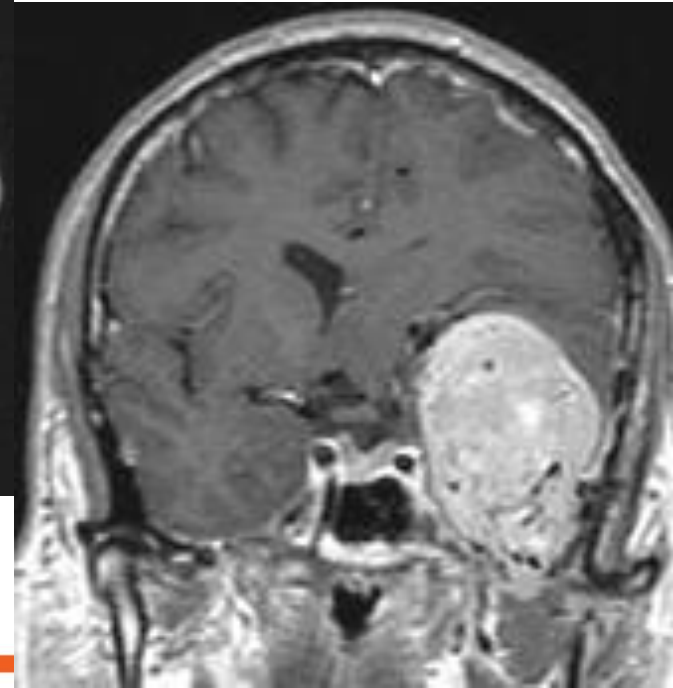
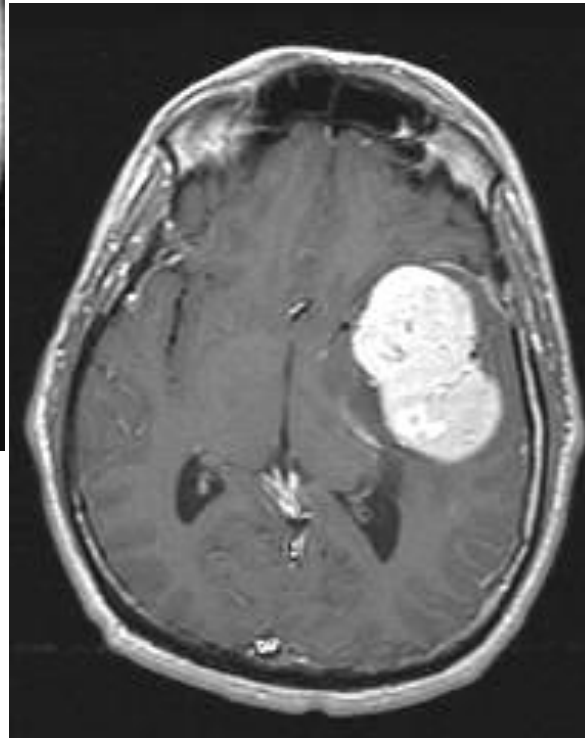
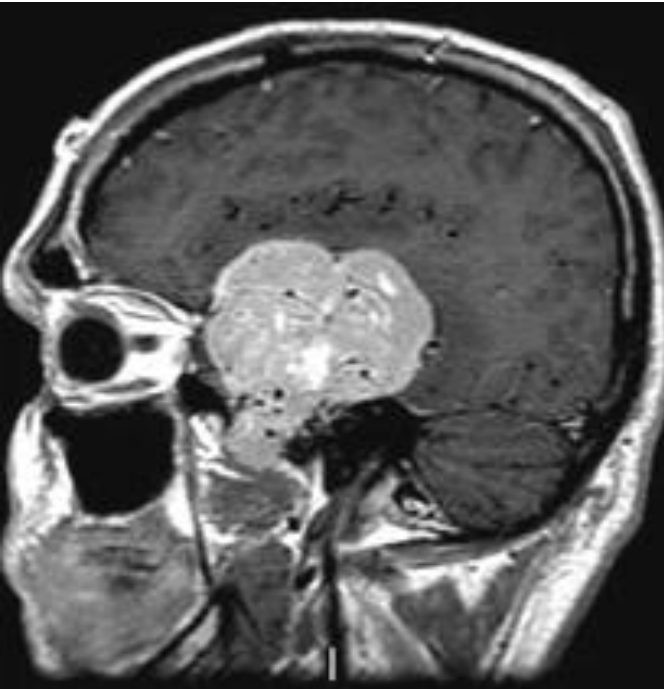
# Imaging Modalities: *CT Scan*

Calcium



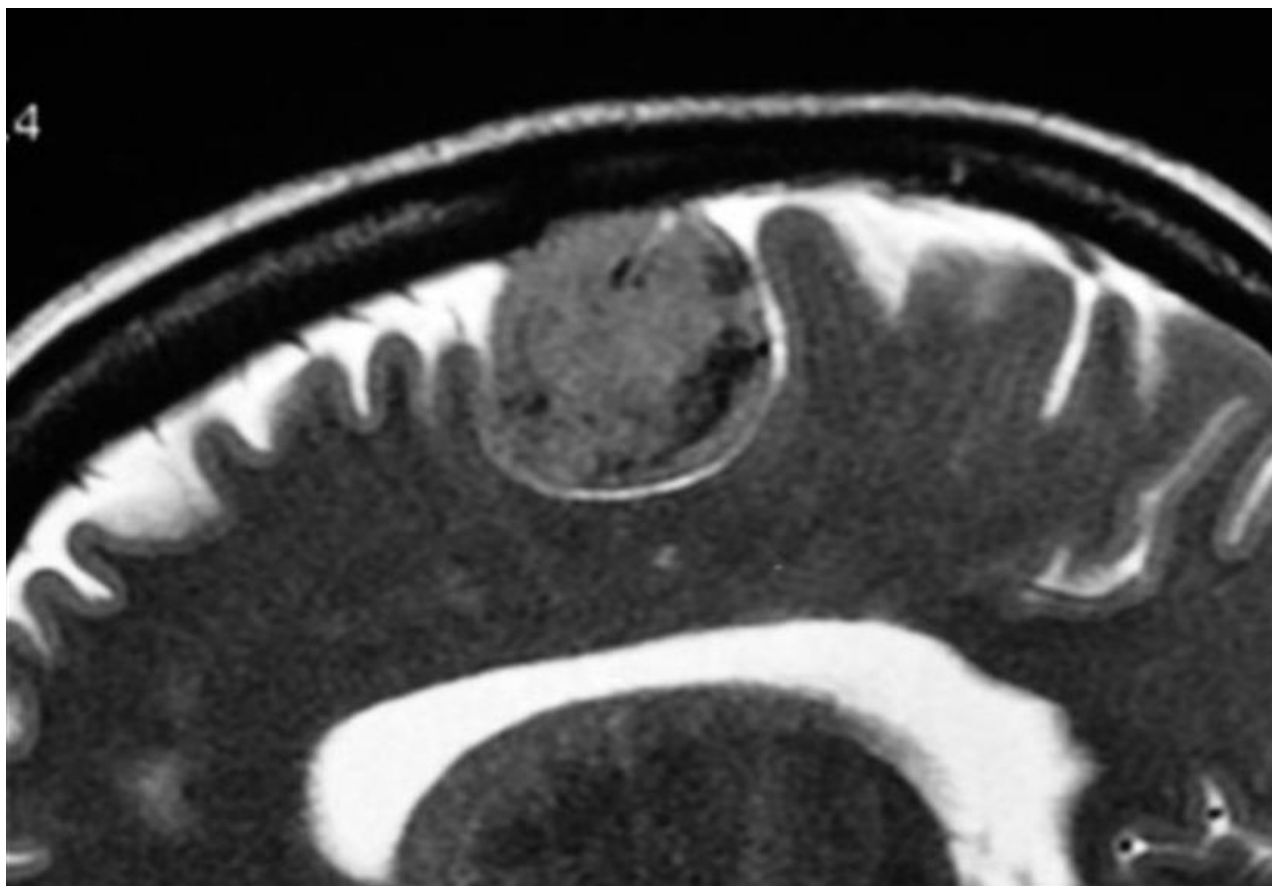
# Imaging Modalities: *MRI Scan*

**Imaging in multiple planes**



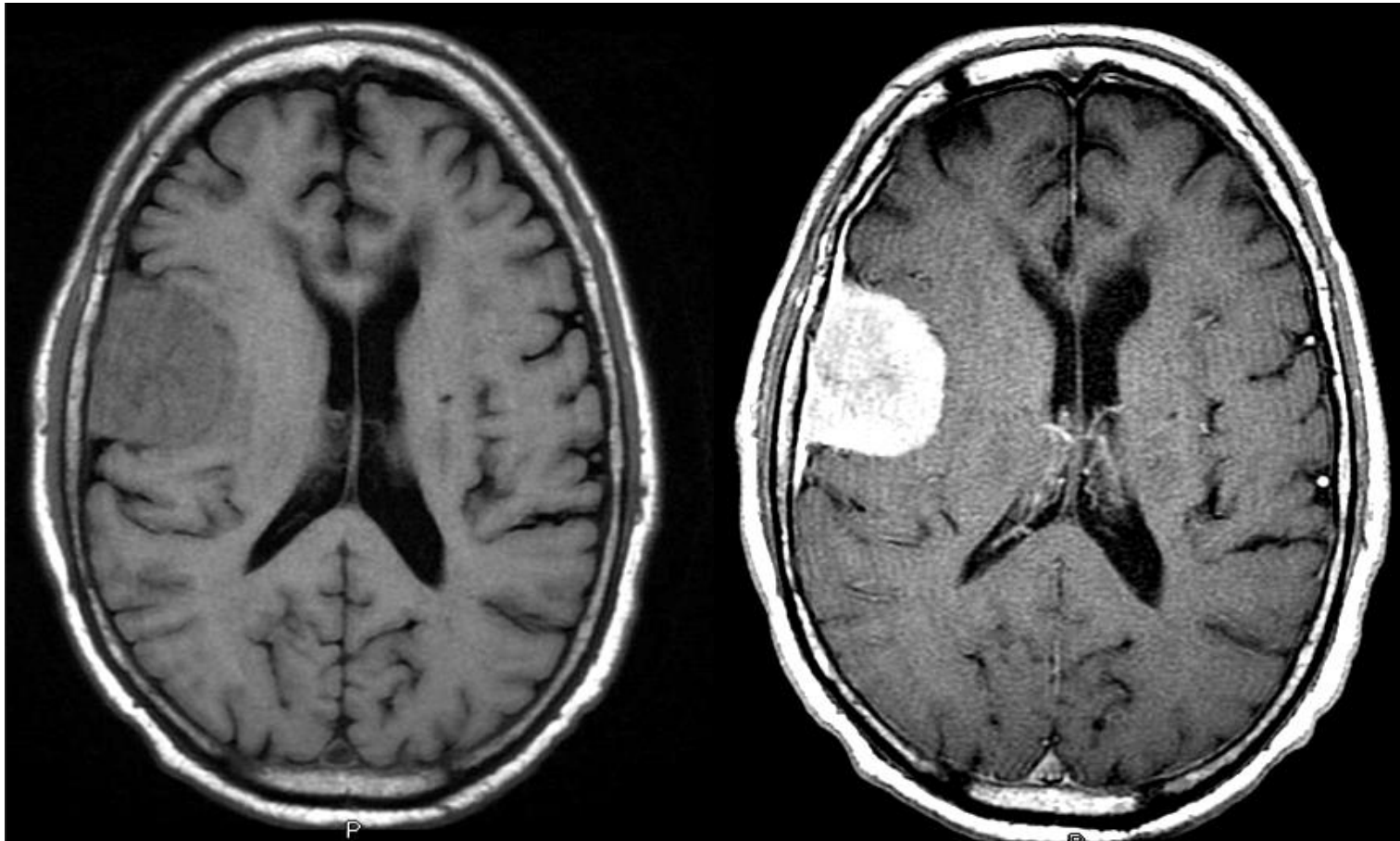
# Imaging Features: *Location*

**Inside the skull but outside the brain**



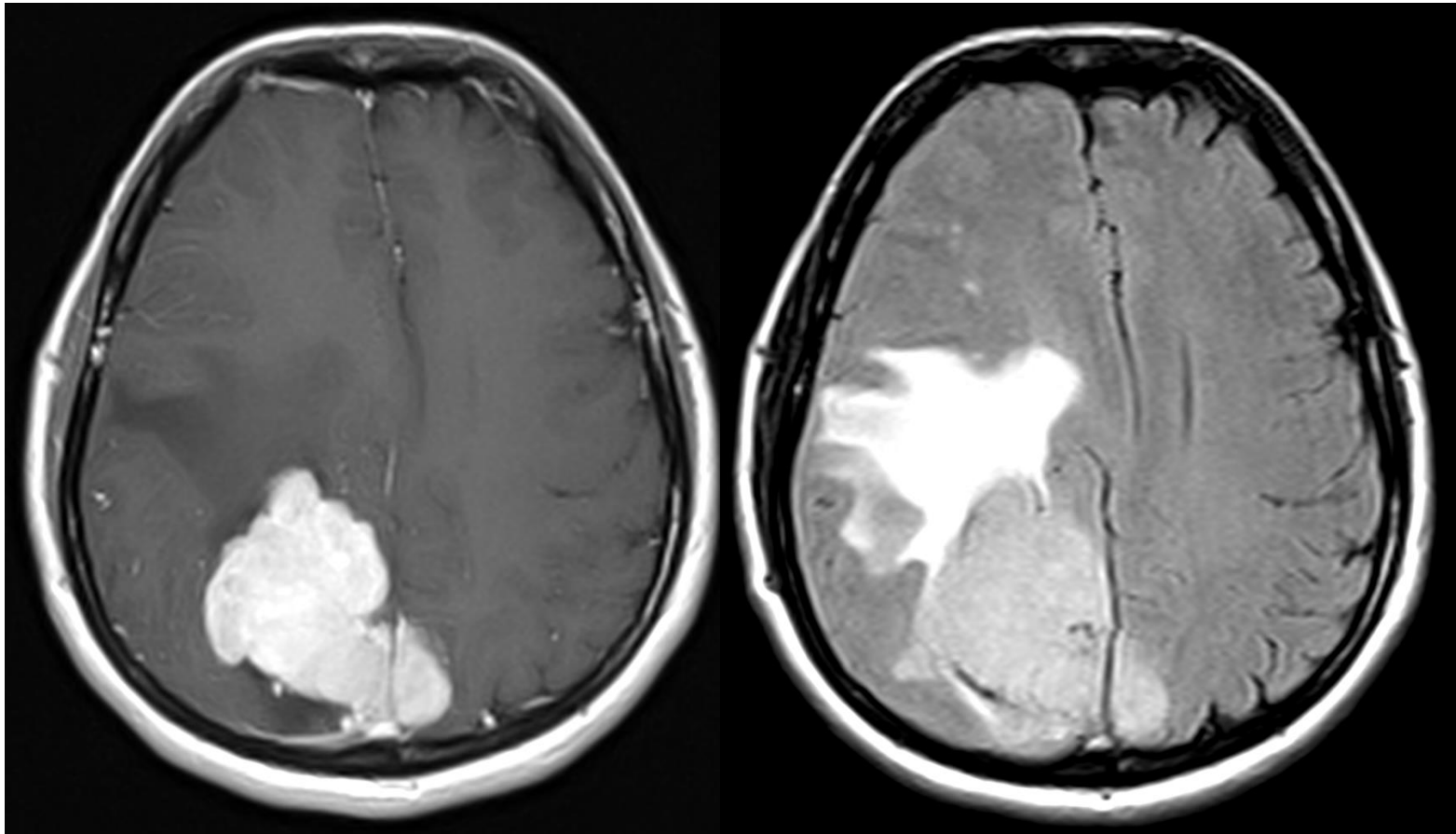
# Imaging Features: *Enhancement*

**Strong enhancement with intravenous dye**



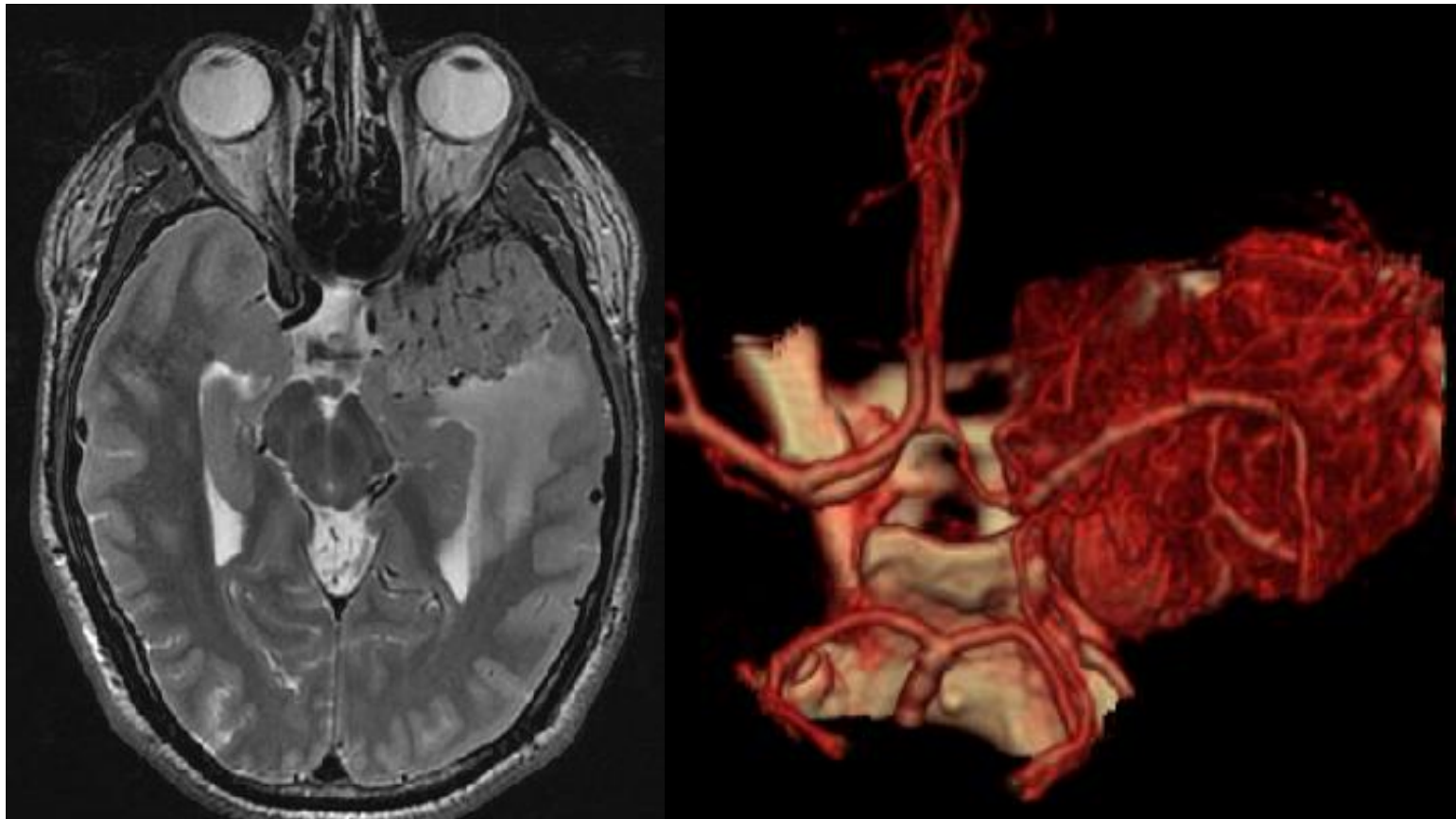
# Imaging Features: *Brain Edema*

**Tumor and surrounding edema  
increase brain pressure**



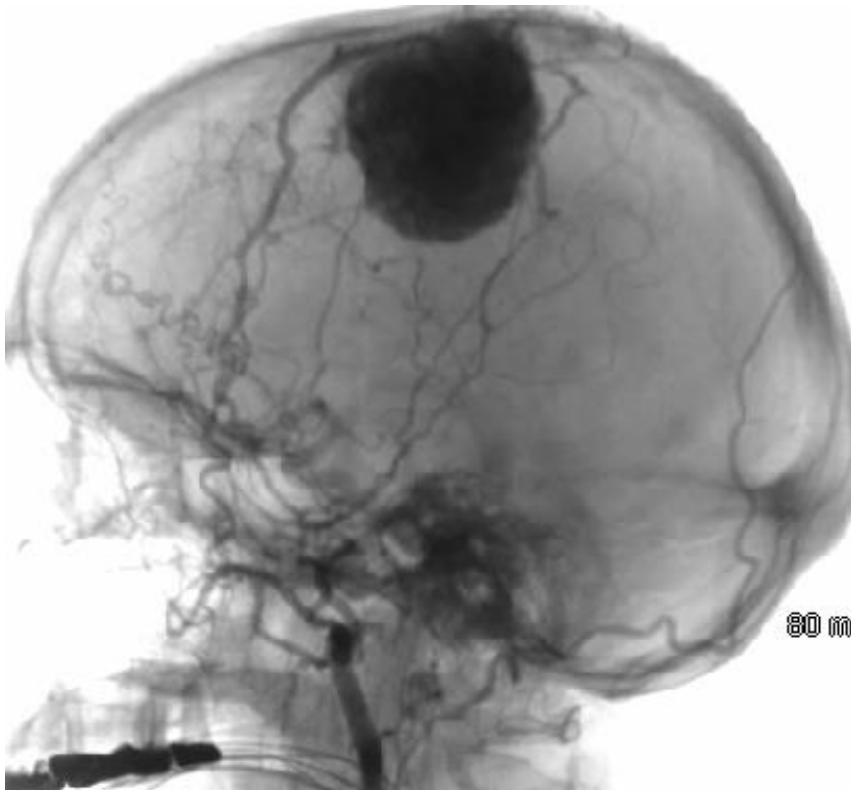
# Imaging Features: *Blood Supply*

**Meningiomas have a rich blood supply**

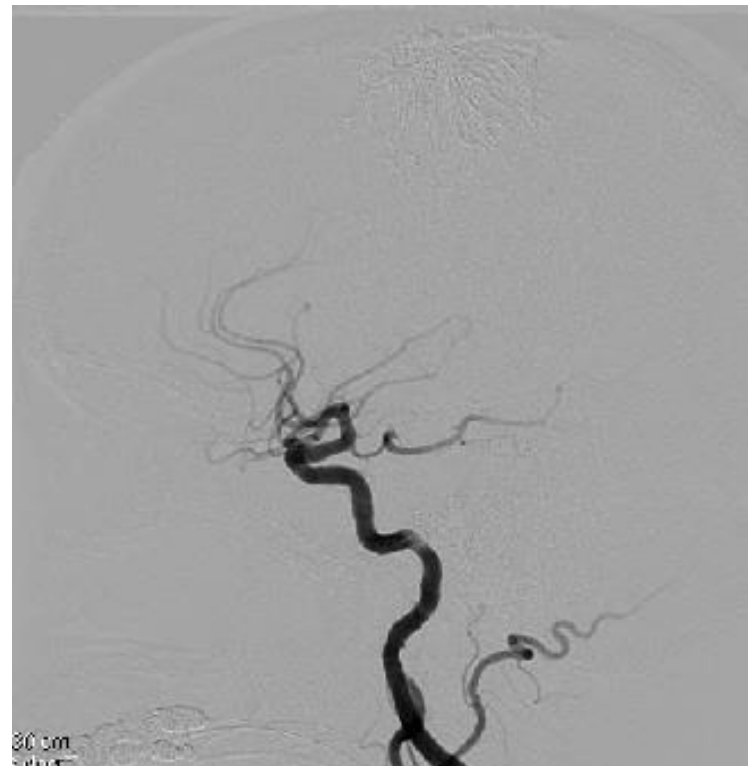


# Angiography and Embolization

**Tumor arteries can be occluded prior to surgery to reduce blood loss**



**Pre-embolization**



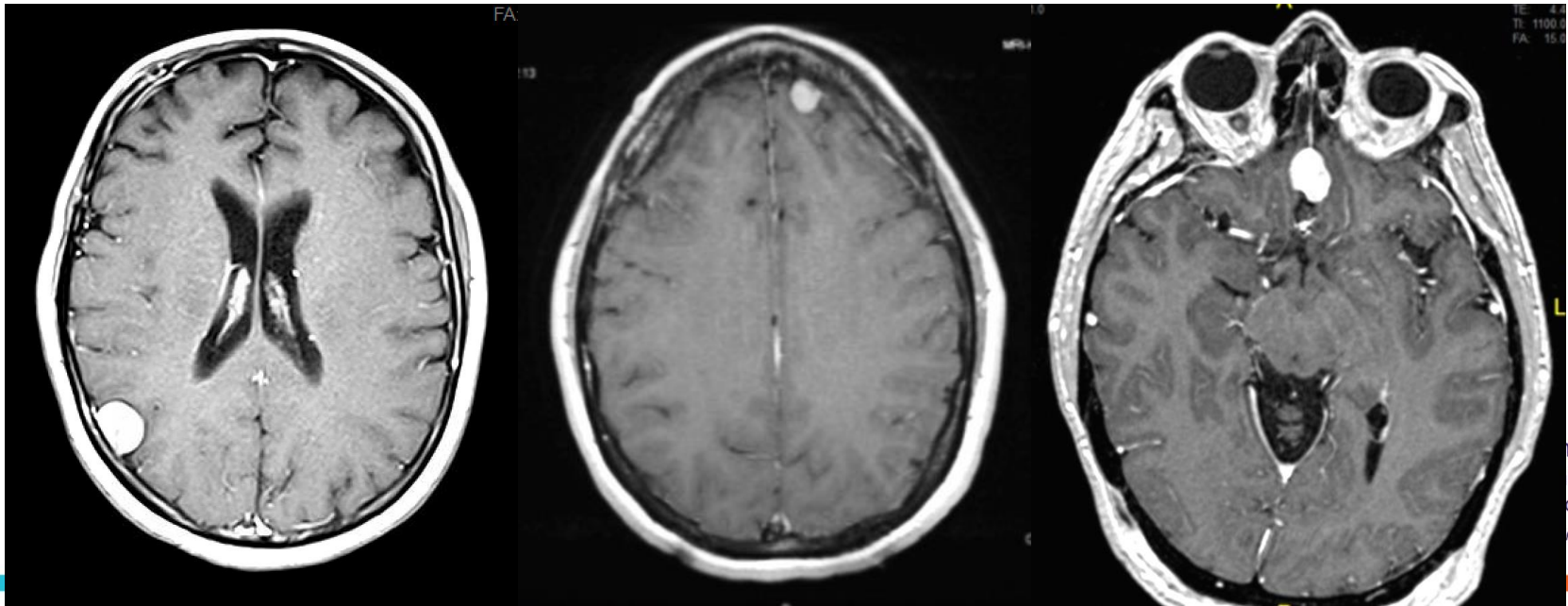
**Post-embolization**

# Treatment Options

- **Observation**
- **Surgery**
- **Radiation**
- **Medical Therapy**

# Observation

- Appropriate for patients w/o symptoms
- Regular MRI scans every 6 – 12 months
- Natural history of meningiomas
  - No change 70%
  - Tumor growth 30%



# Rationale for Surgery

## ■ Pathology

- Confirms diagnosis and grade of tumor
- Special studies (hormone receptors)

## ■ Neurologic

- Improves neurological status
- Prevents future deterioration

## ■ Outcome

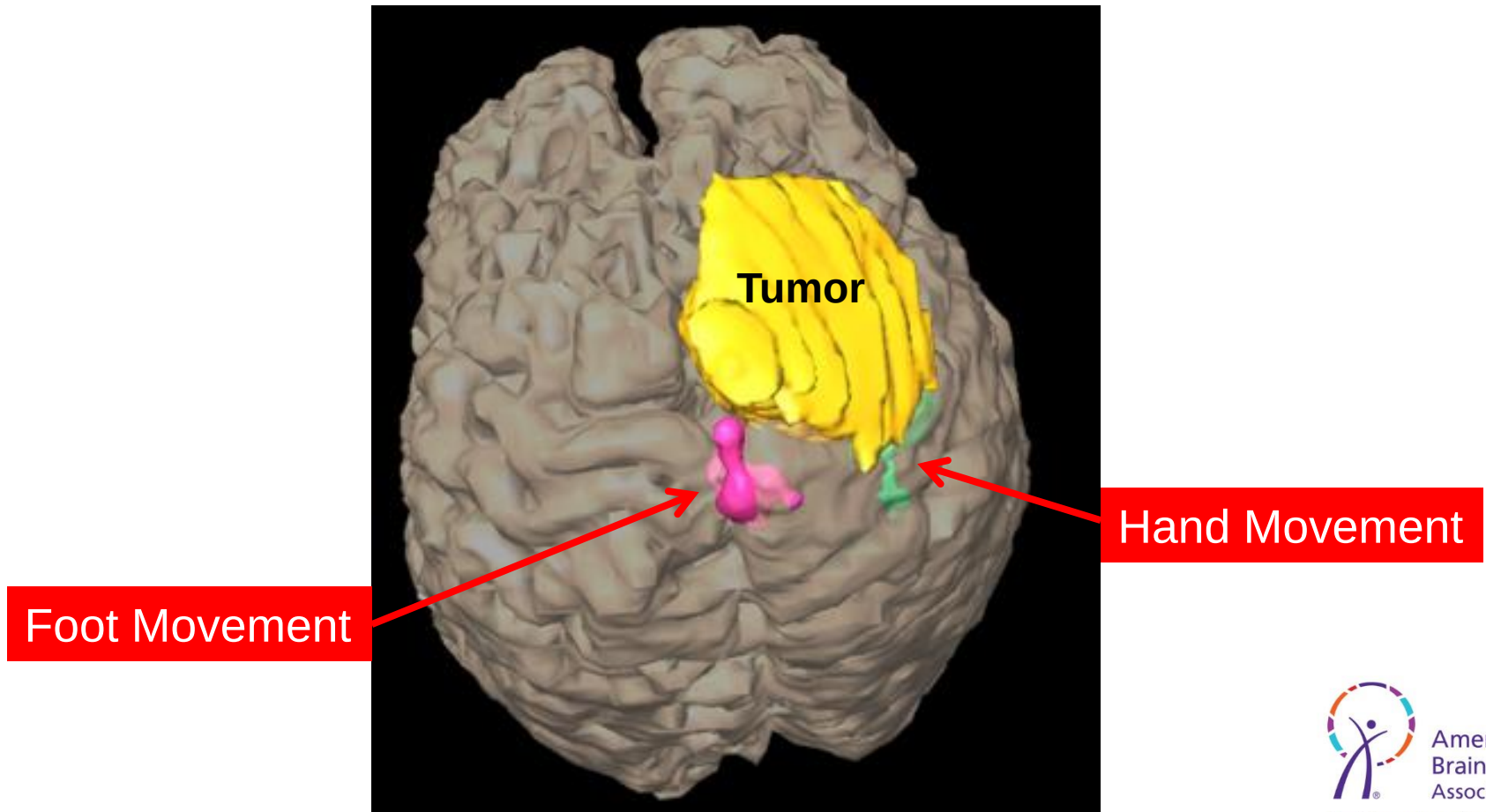
- Complete removal can be curative

# What Factors Influence the Decision to Undergo Surgery?

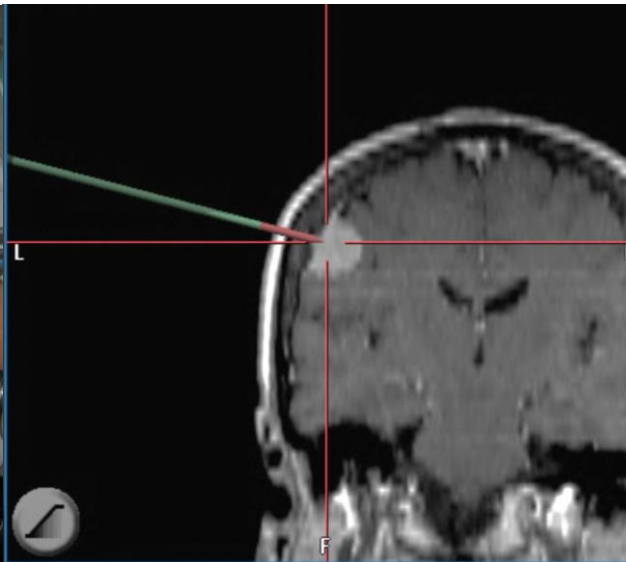
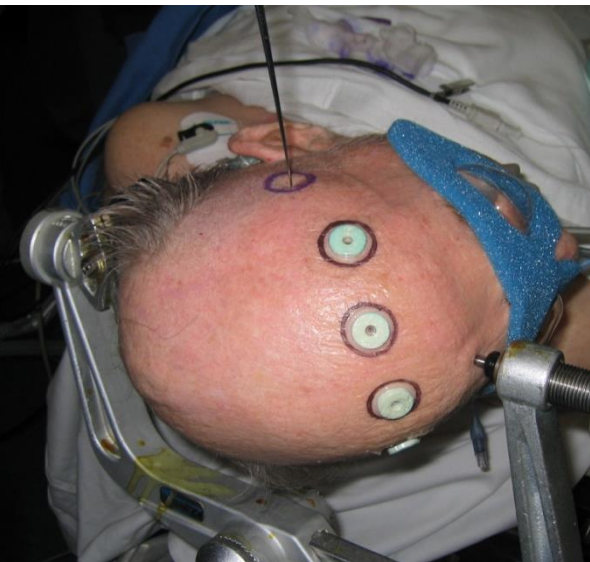
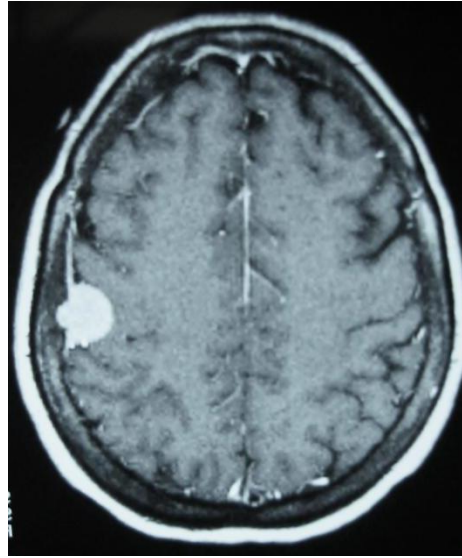
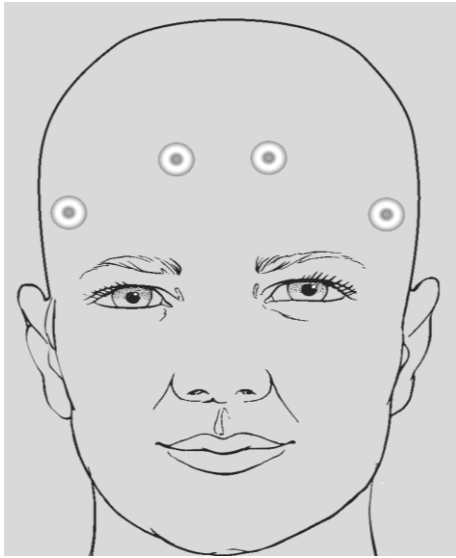
| <b>Tumor</b>          | <b>Patient</b>           | <b>Surgeon</b>    |
|-----------------------|--------------------------|-------------------|
| <b>Location</b>       | <b>Age</b>               | <b>Ability</b>    |
| <b>Size</b>           | <b>Neurologic status</b> | <b>Experience</b> |
| <b>Brain swelling</b> | <b>Medical condition</b> | <b>Resources</b>  |
| <b>Eloquent areas</b> | <b>Patient choice</b>    |                   |
| <b>Invasiveness</b>   |                          |                   |
| <b>Presumed grade</b> |                          |                   |

# Surgical Pre-Planning

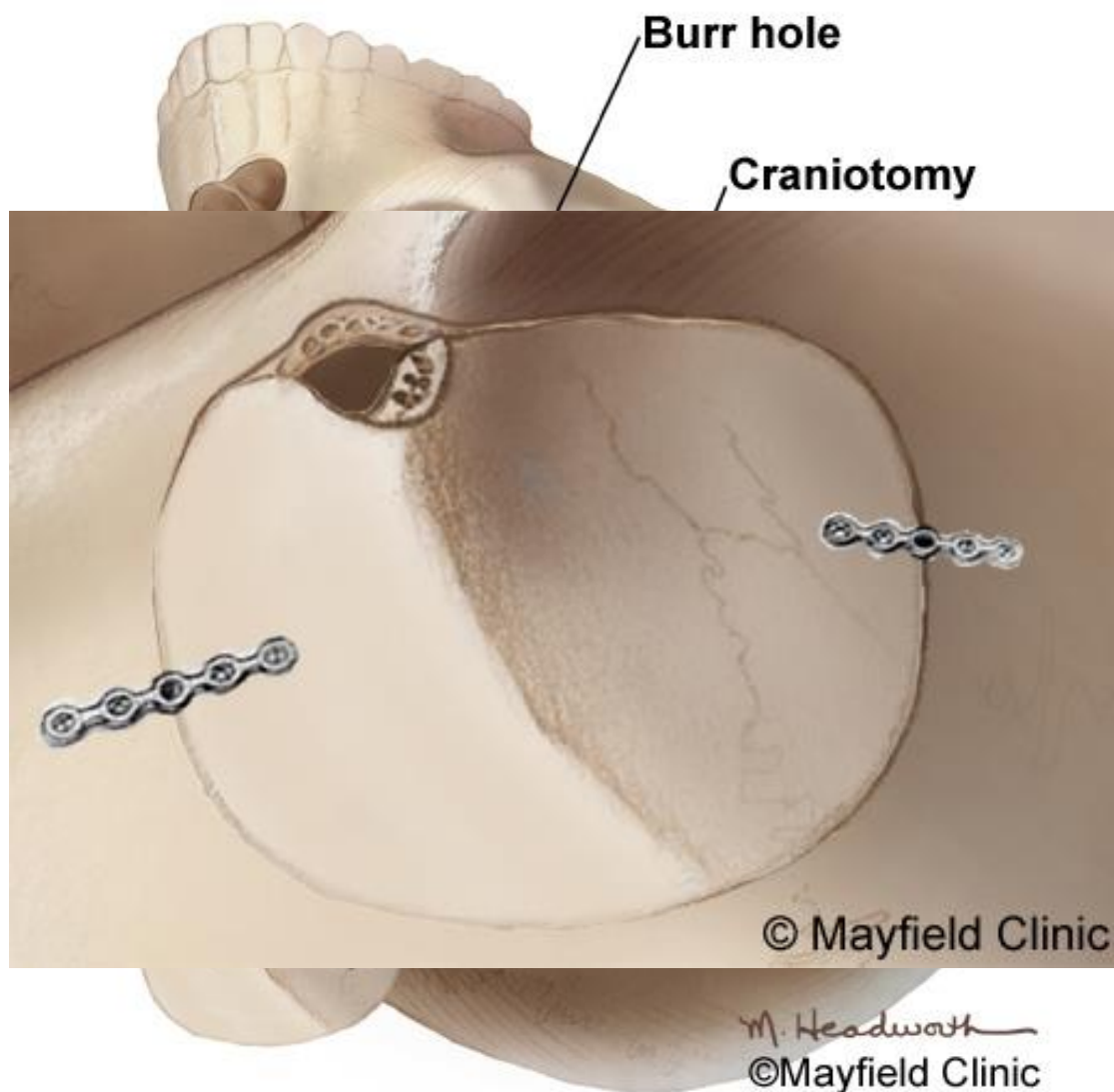
Provides a preoperative roadmap for the neurosurgeon



# Computer-Guided Surgery



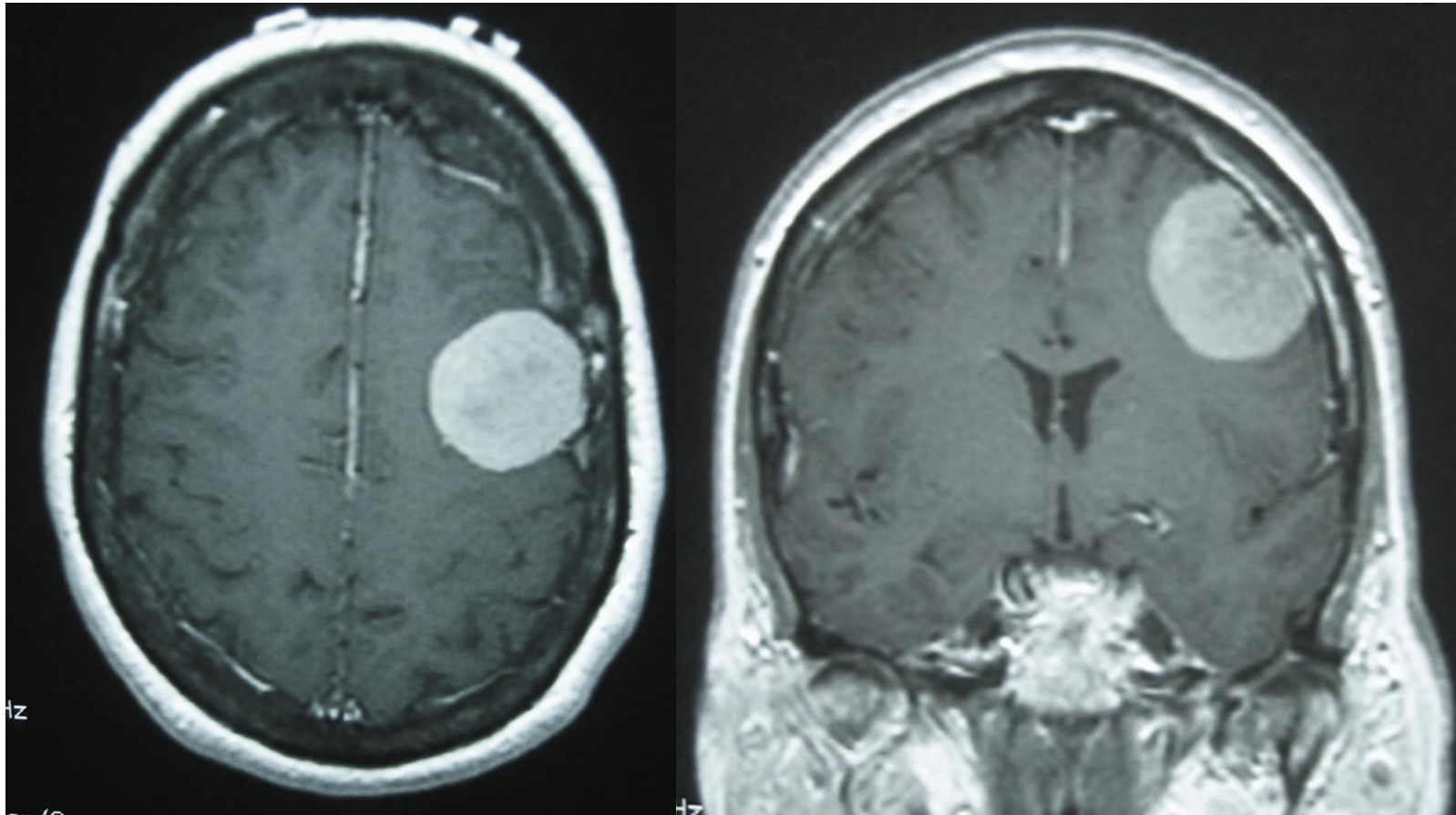
# Craniotomy for Meningioma Removal



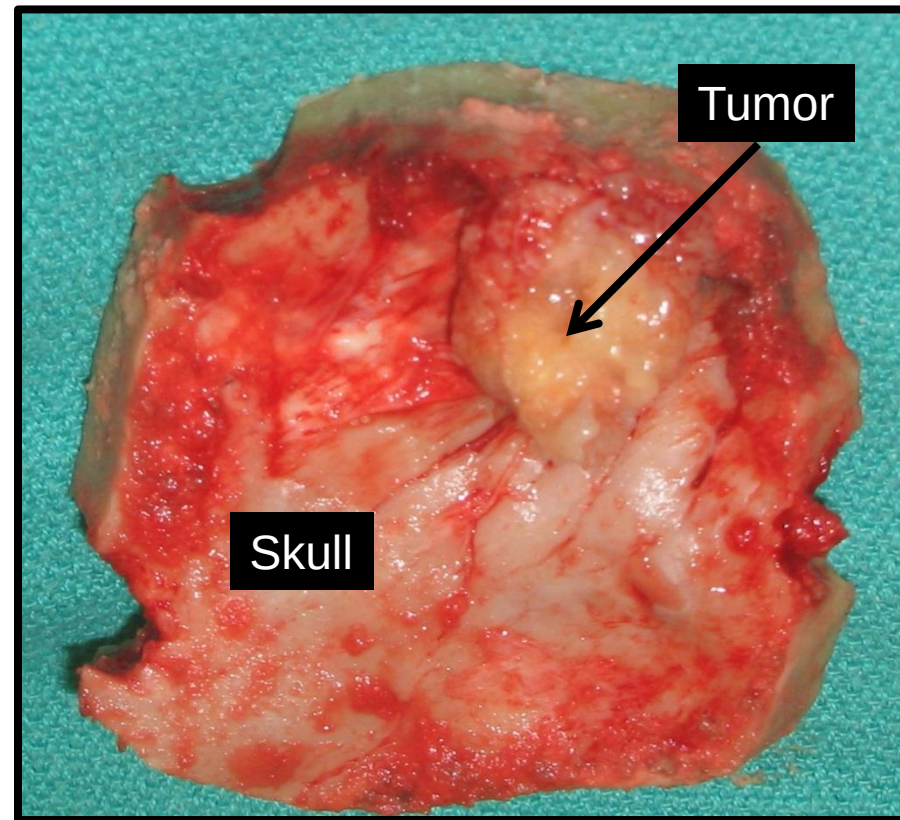
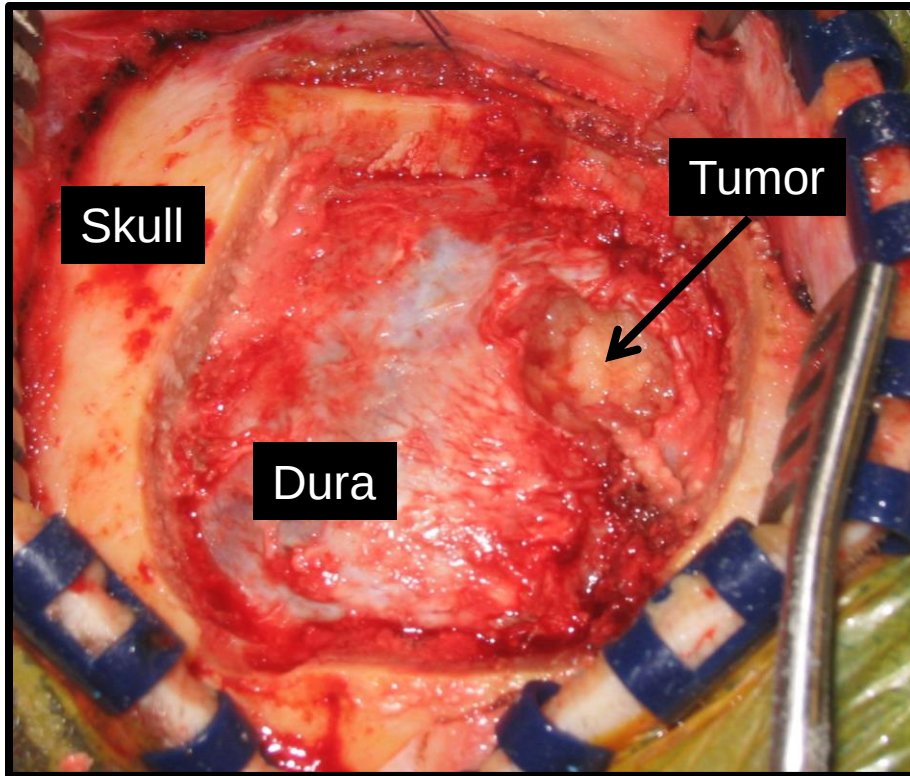
# Meningioma Surgery Case Study

- **59 year-old woman with a long history of migraines headaches. Headaches increased in frequency and severity which prompted an MRI scan.**
- **Neurologic Exam: normal**

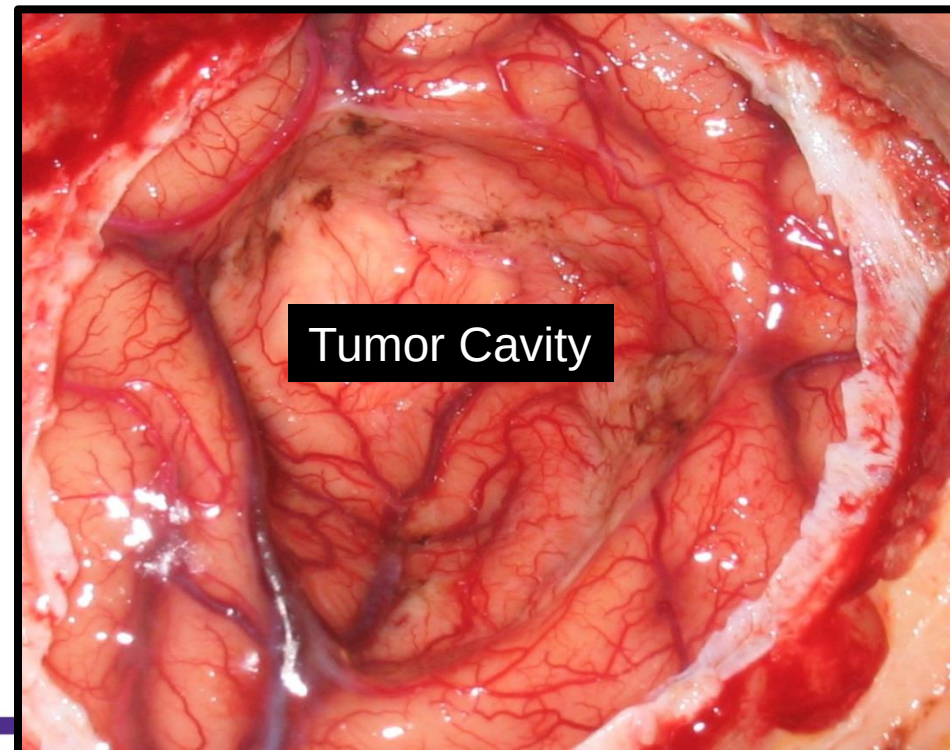
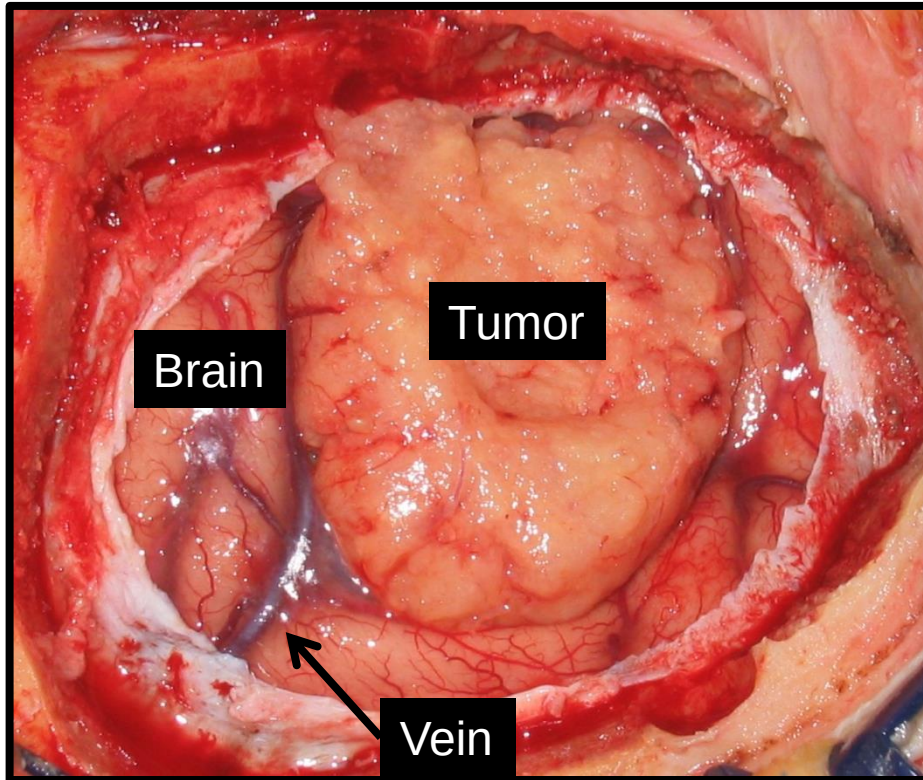
# Preoperative MRI



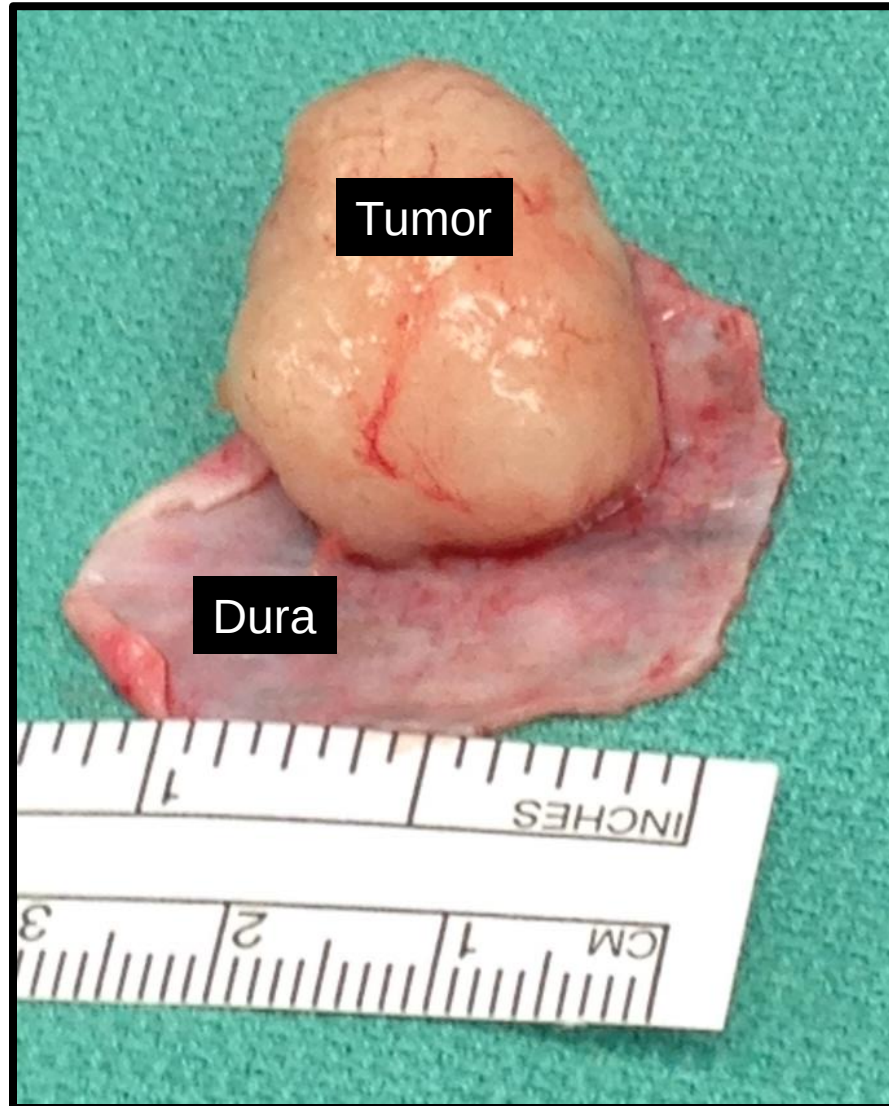
# Meningioma Exposure



# Meningioma Removal

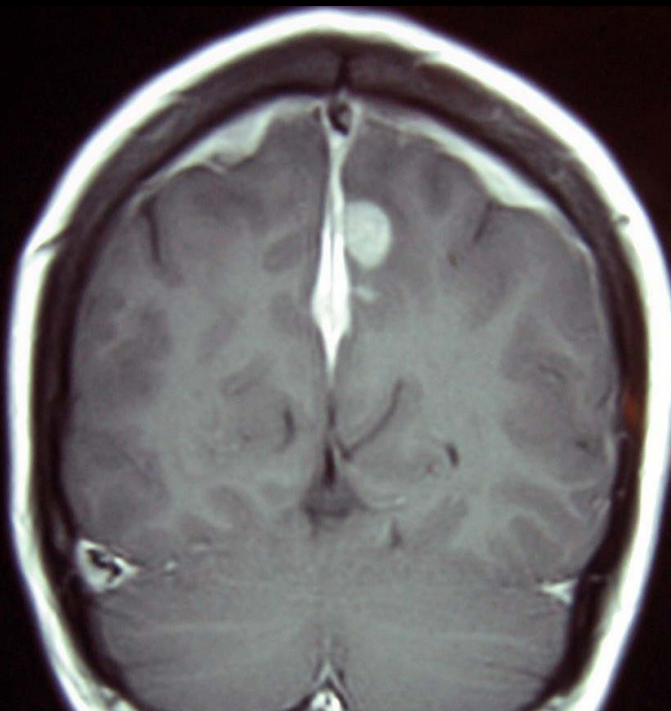
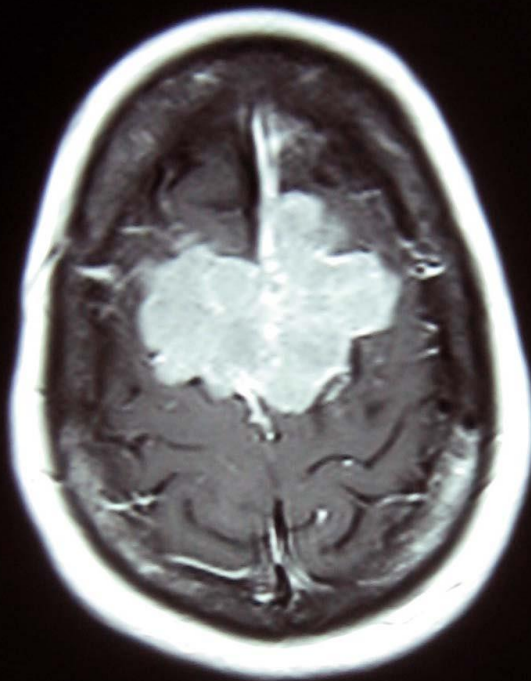
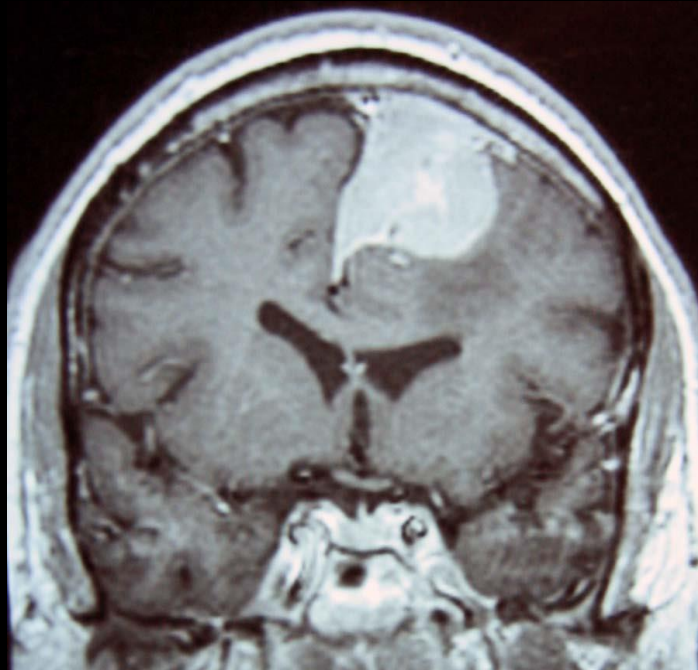
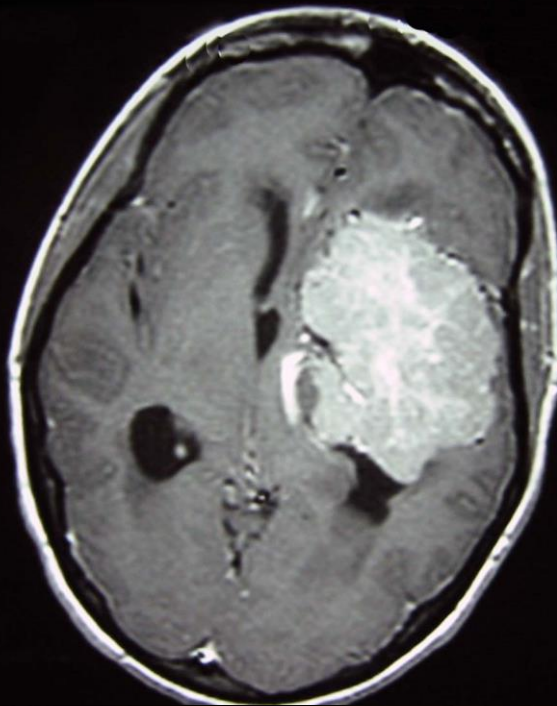


# Meningioma Specimen



# Limitations of Surgery

- **Vascular structures**
  - Arteries
  - Veins
- **Eloquent areas**
  - Optic nerve
  - Cranial nerves
  - Motor/Sensory/Language/Visual
- **Highly Invasive or Multiple Meningiomas**



# Risks of Surgery

- **Neurologic**

- Bleeding
- Brain swelling
- Seizures
- Stroke

- **Regional**

- Wound infection
- Wound breakdown

- **General**

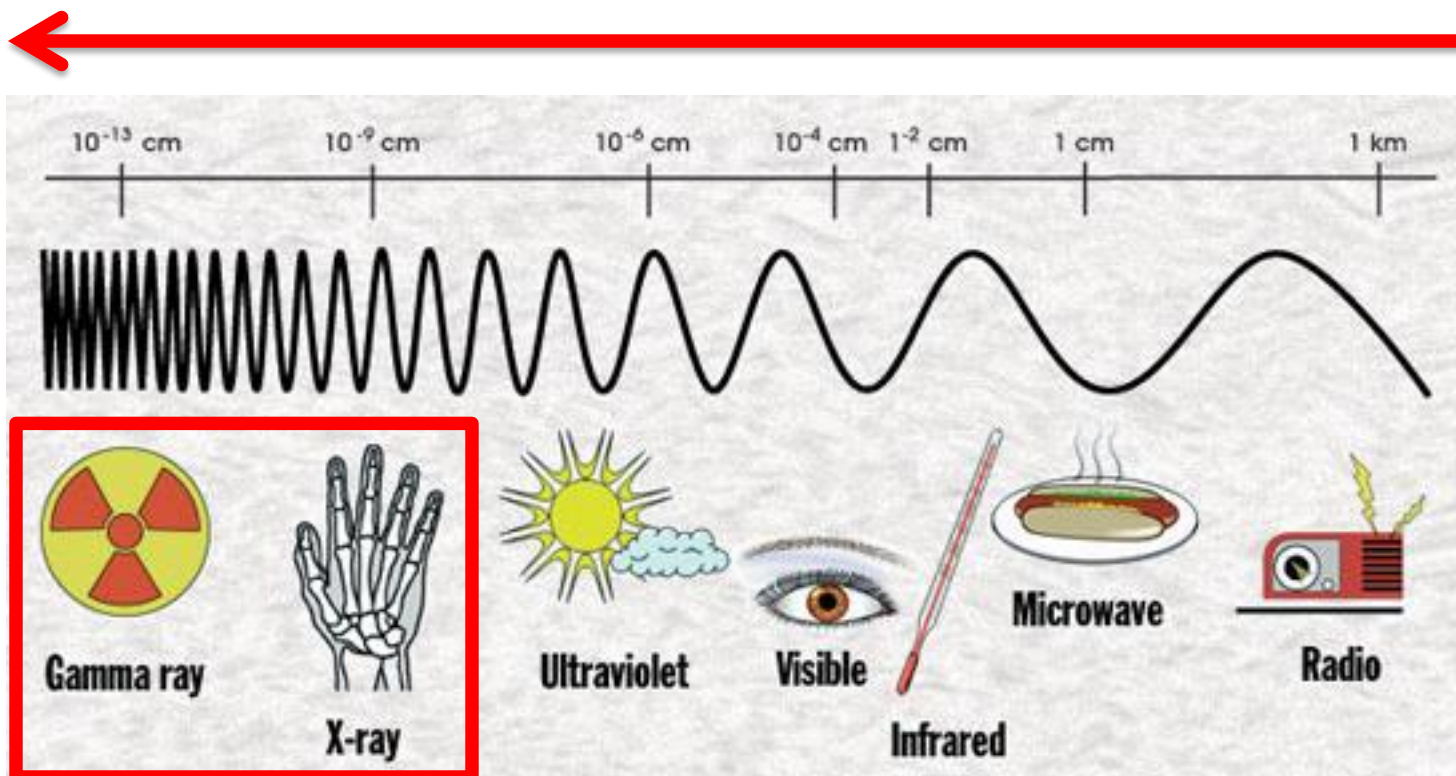
- Anesthetic complications
- Heart/Lungs
- Urine infection
- Blood clots

# Types of Radiation

High

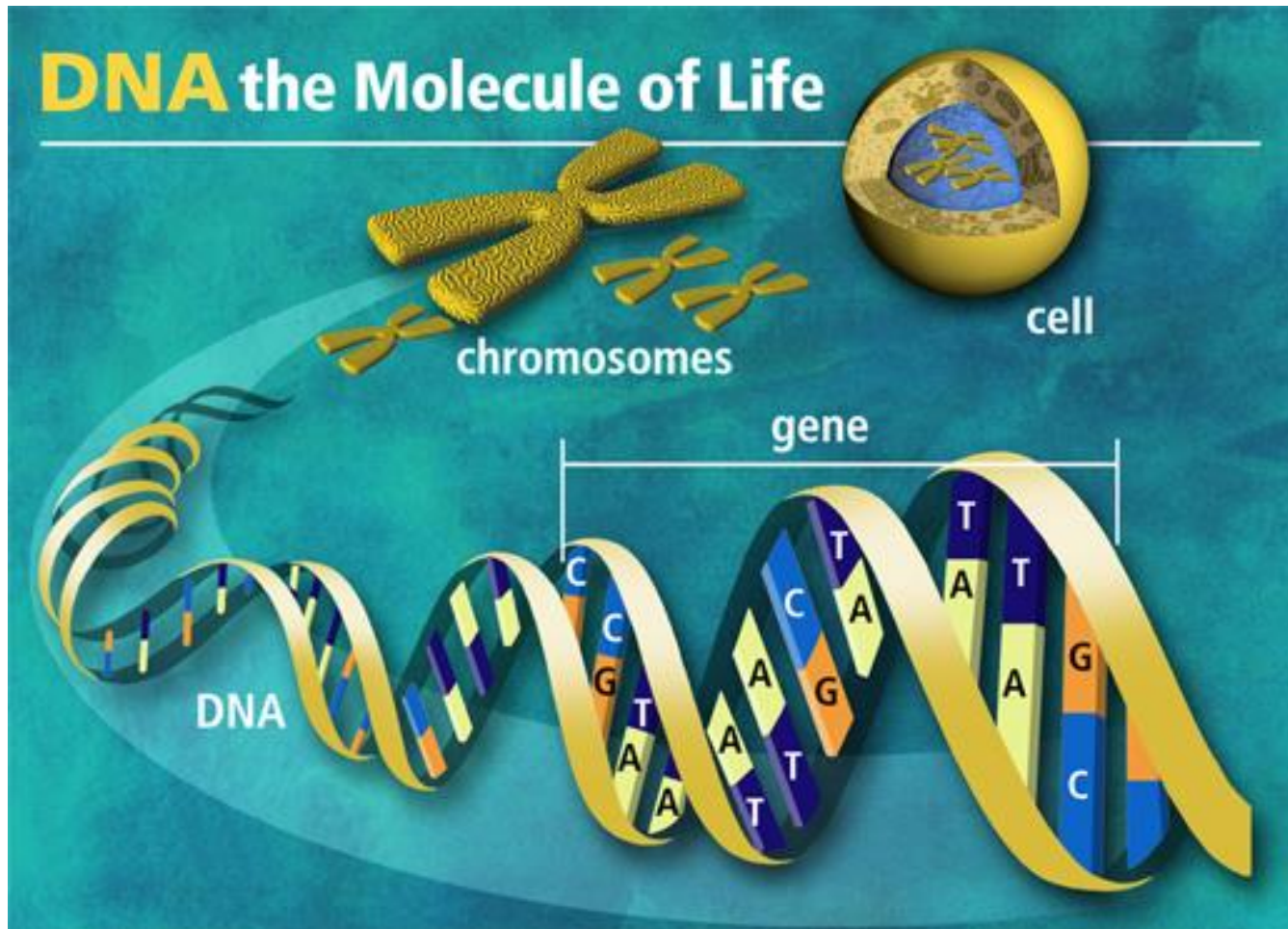
ENERGY LEVEL

Low

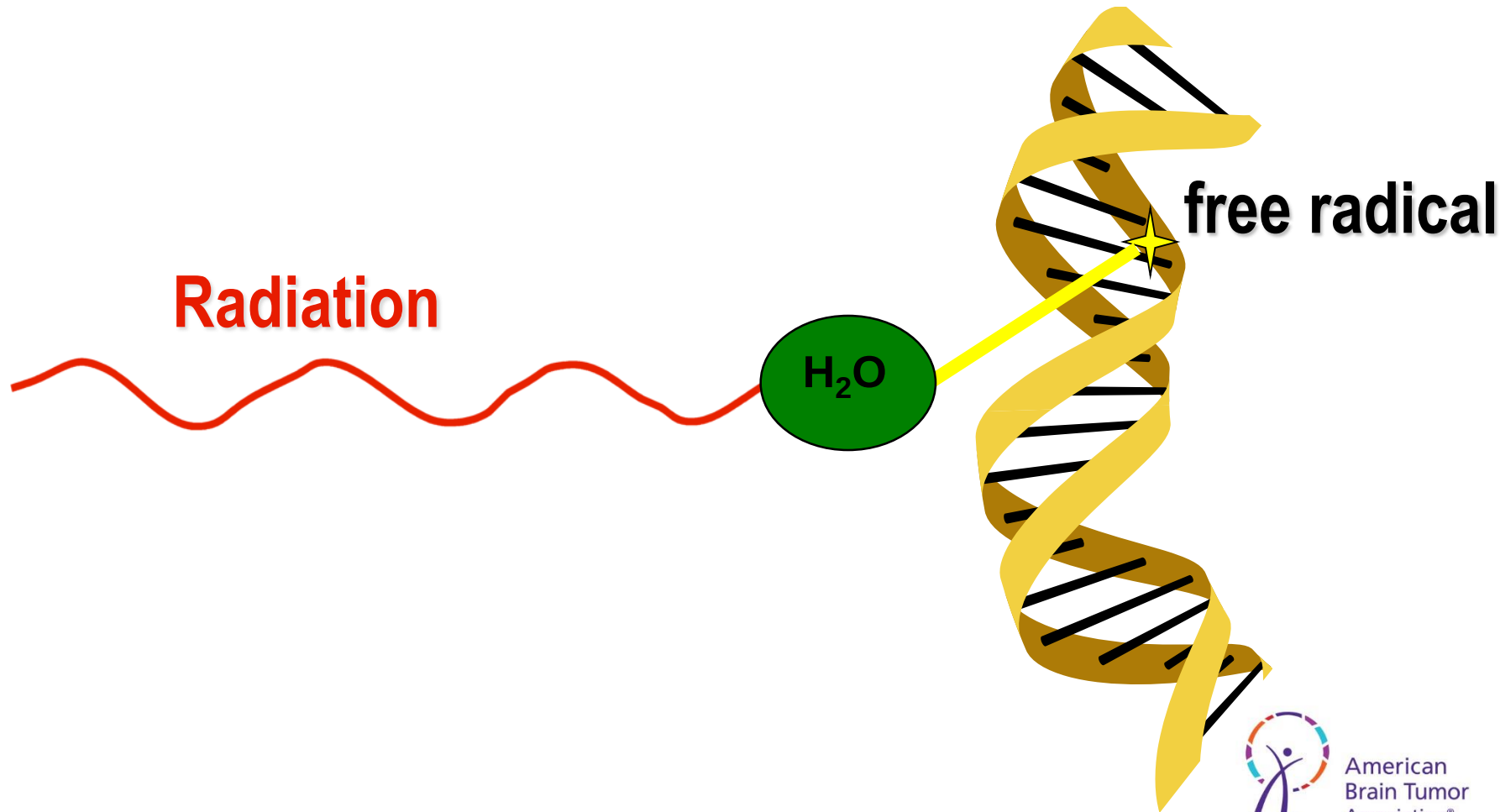


**Meningioma Radiation**

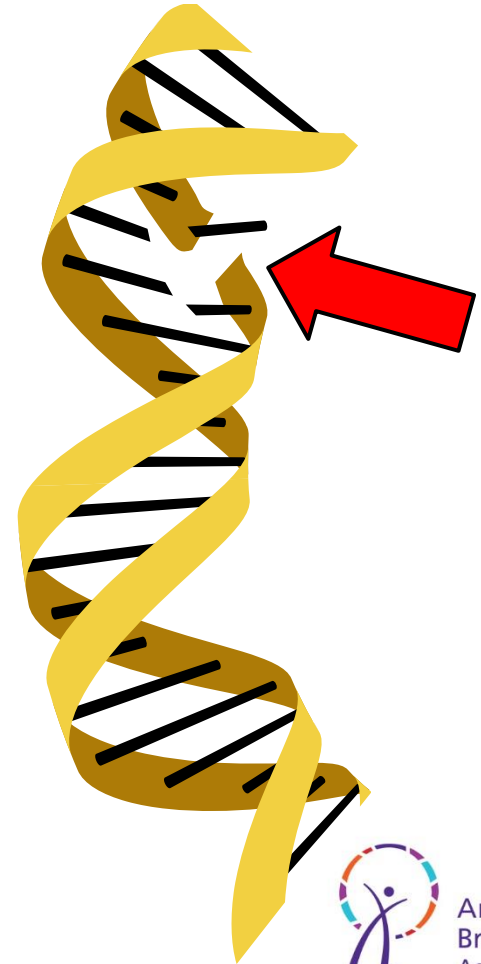
# Tumor DNA is the Target of Radiation



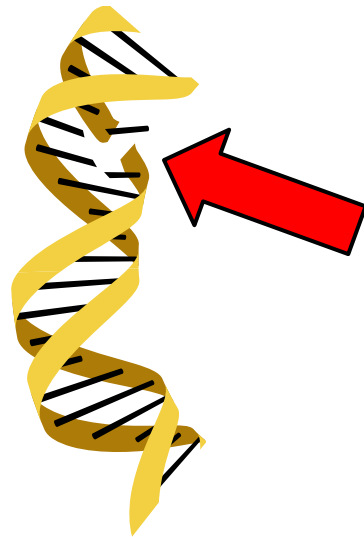
# Radiation delivers energy to the tumor cell DNA



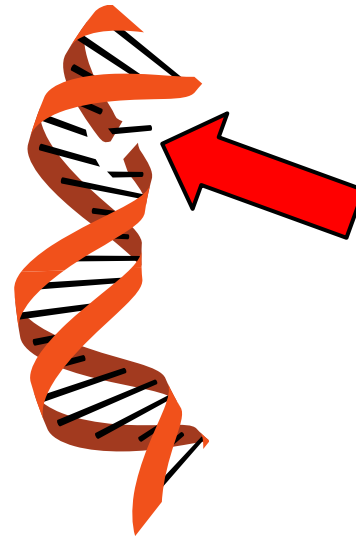
# Which damages the DNA



# Radiation damages both normal and tumor cells



**Normal brain cell**



**Tumor cell**

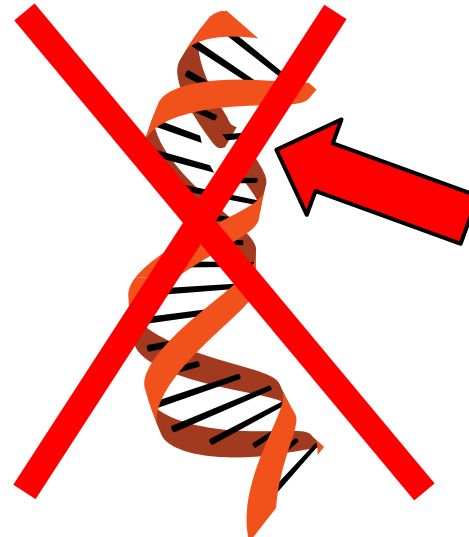
# Tumor cells are unable to repair damage (normal brain cells are able)

**Repaired**



**Normal brain cell**

**Unrepaired**



**Tumor cell**

# Common Radiation Machines

True Beam



Novalis



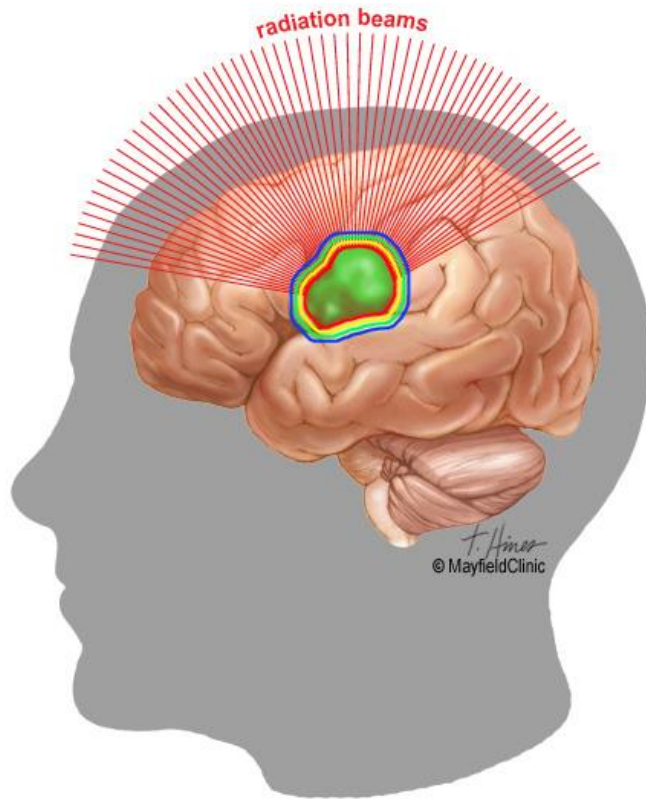
Cyberknife



Gamma Knife

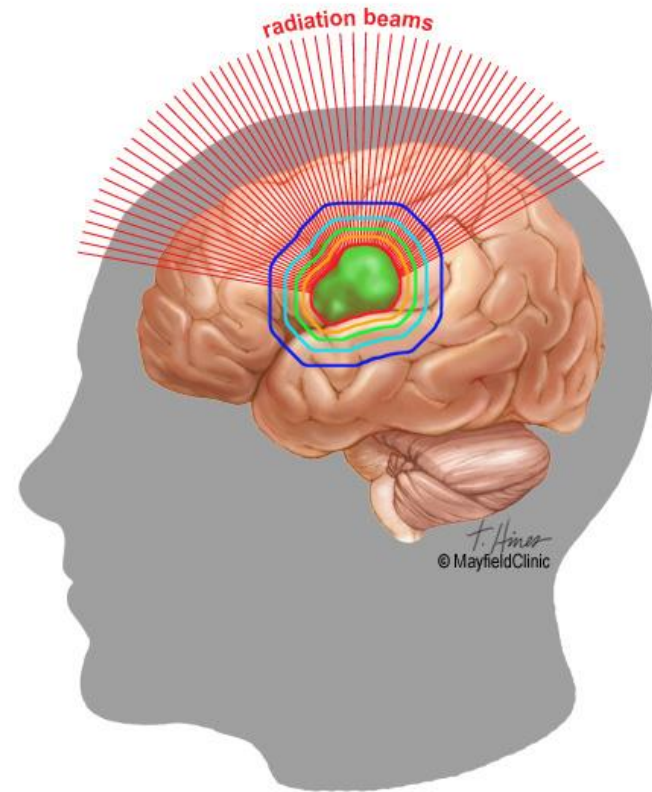


# Radiosurgery



- High dose
- 1-5 sessions
- Tight treatment area

# Radiotherapy

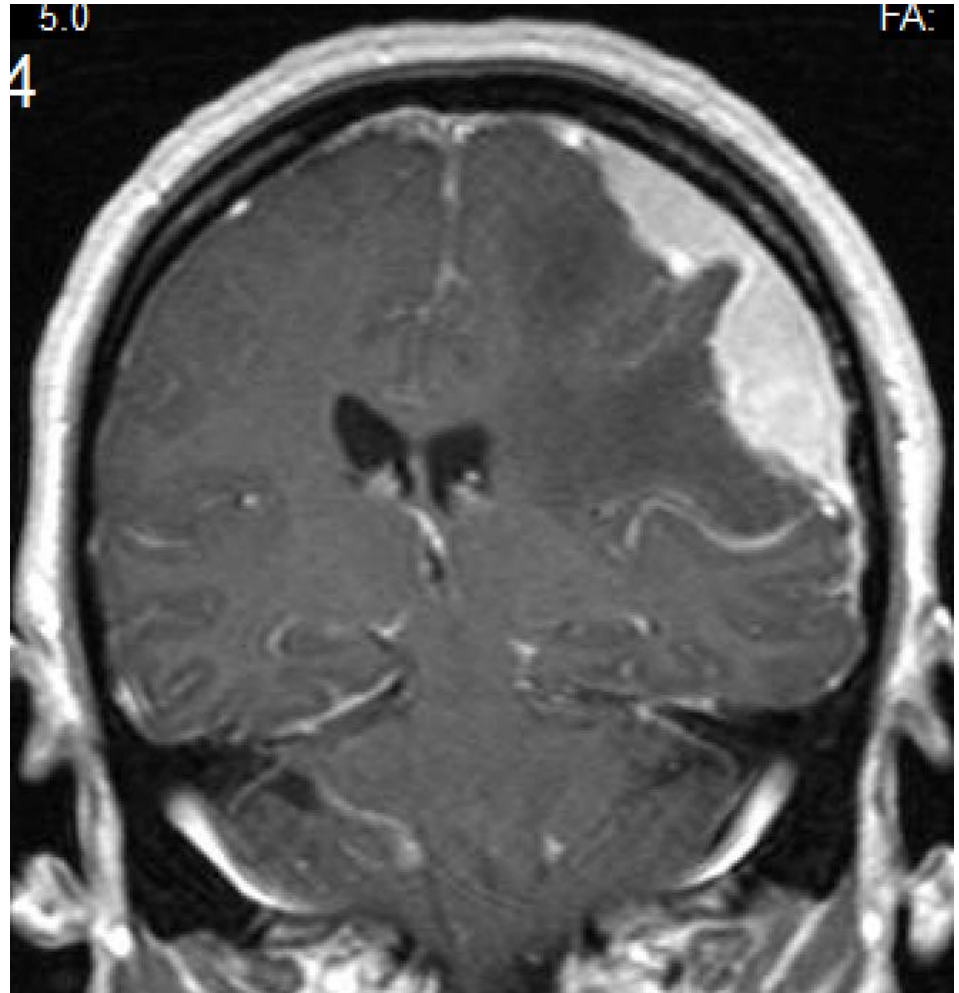


- Low dose
- 6-33 sessions
- Margin of normal brain

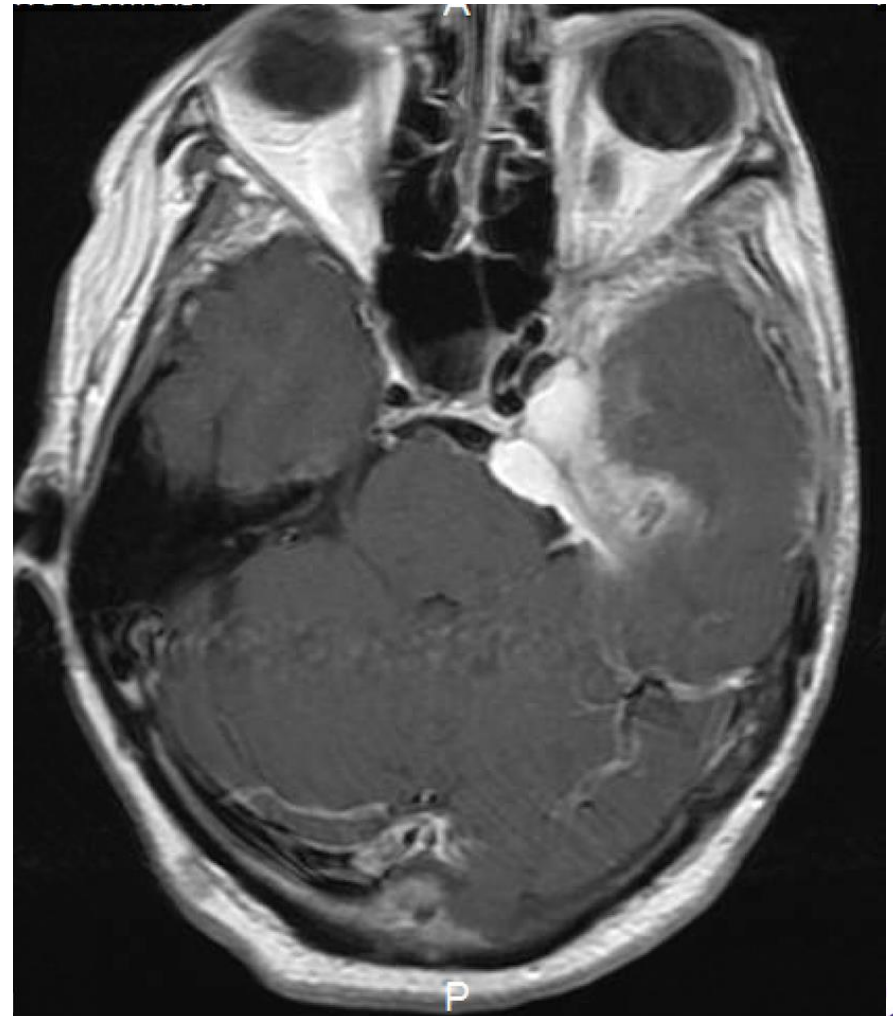
# Indications for Radiation

- Inoperable meningiomas
- After surgery for incompletely removed meningiomas
- After surgery for malignant meningioma
- Recurrent tumors after previous surgery
- Co-existing medical problems that preclude surgery
- Patient choice

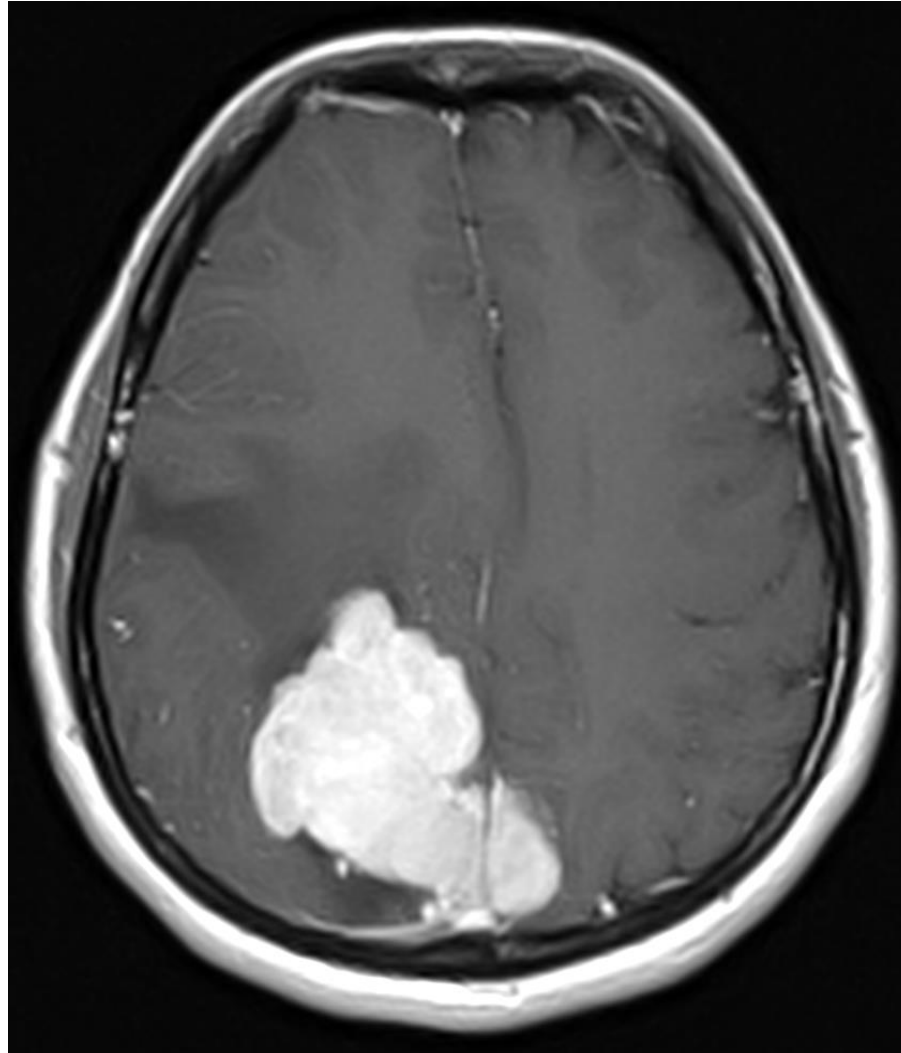
# Inoperable



# Incomplete Removal

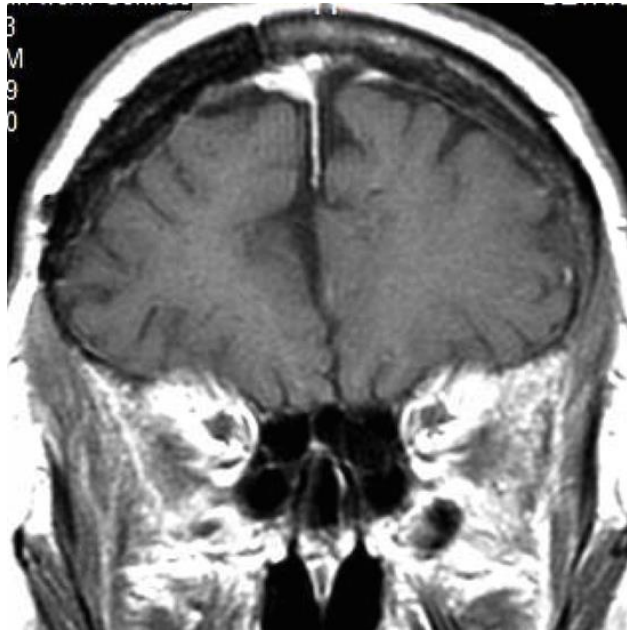


# Malignant Meningioma

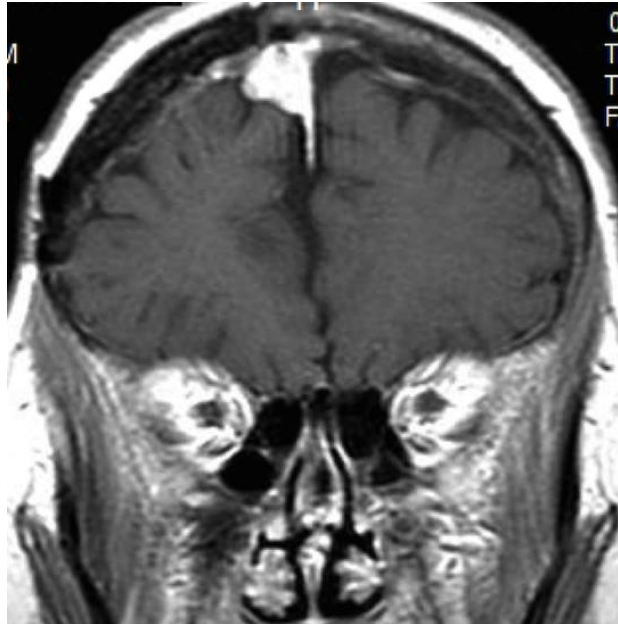


# Recurrent Meningioma

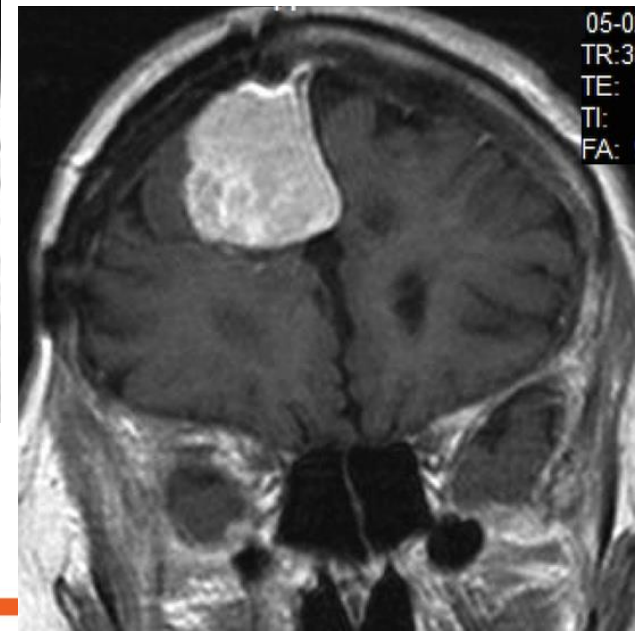
2003



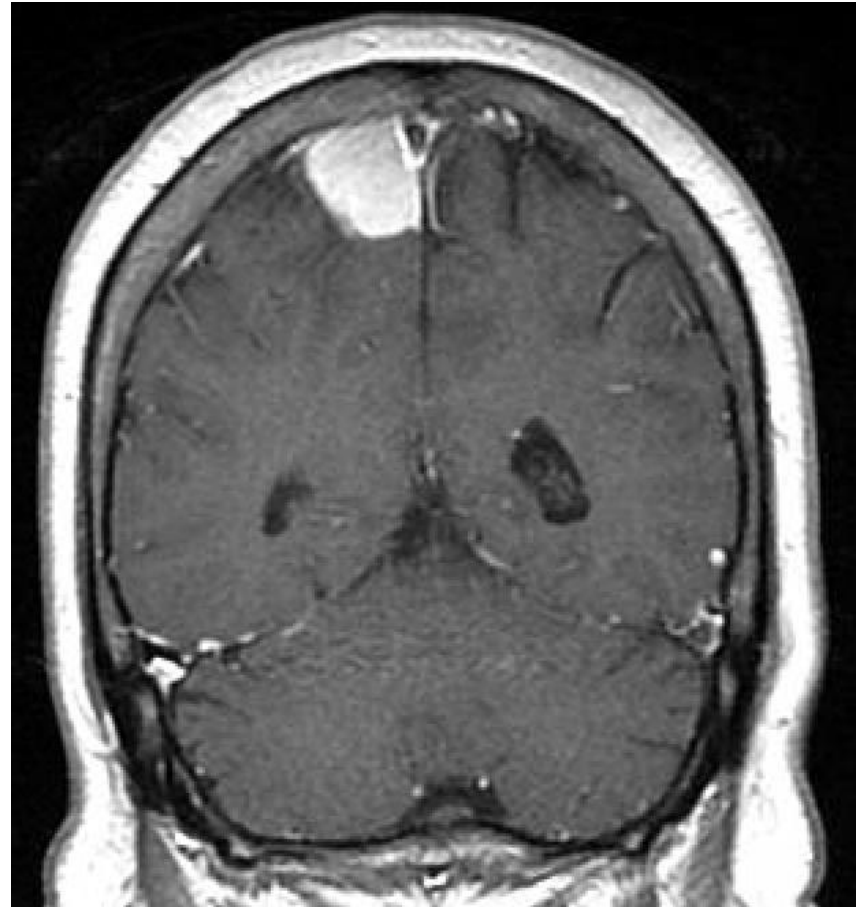
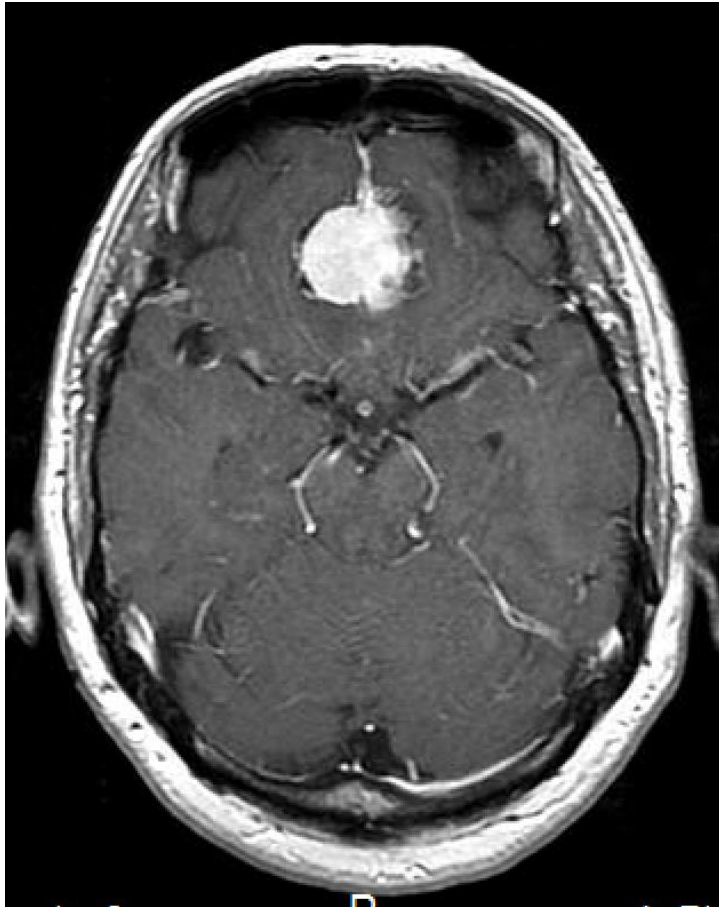
2005



2014



# Co-existing Medical Problems or Patient Choice

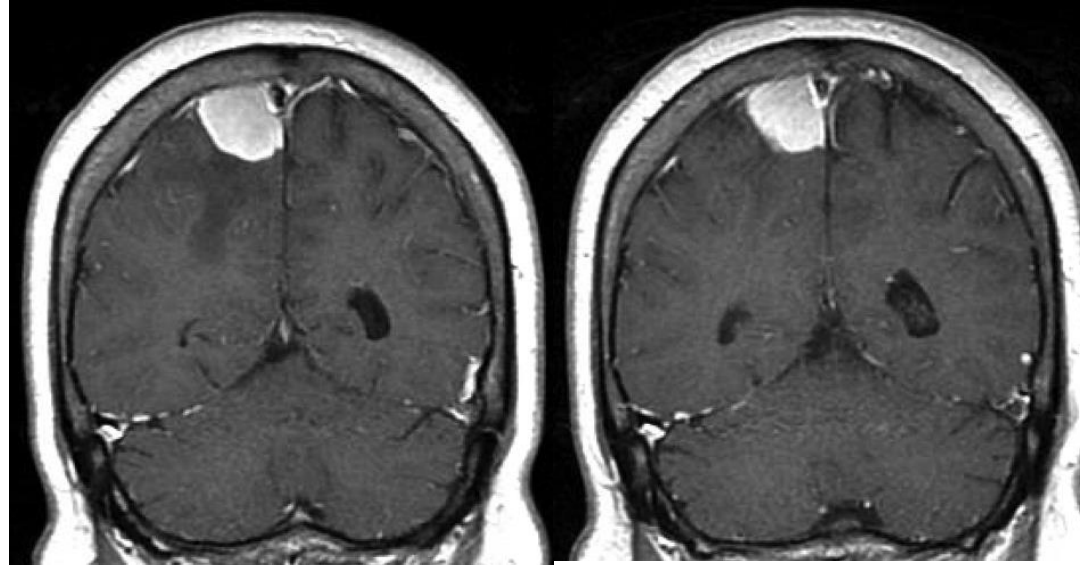


# Results of Radiation for Meningioma

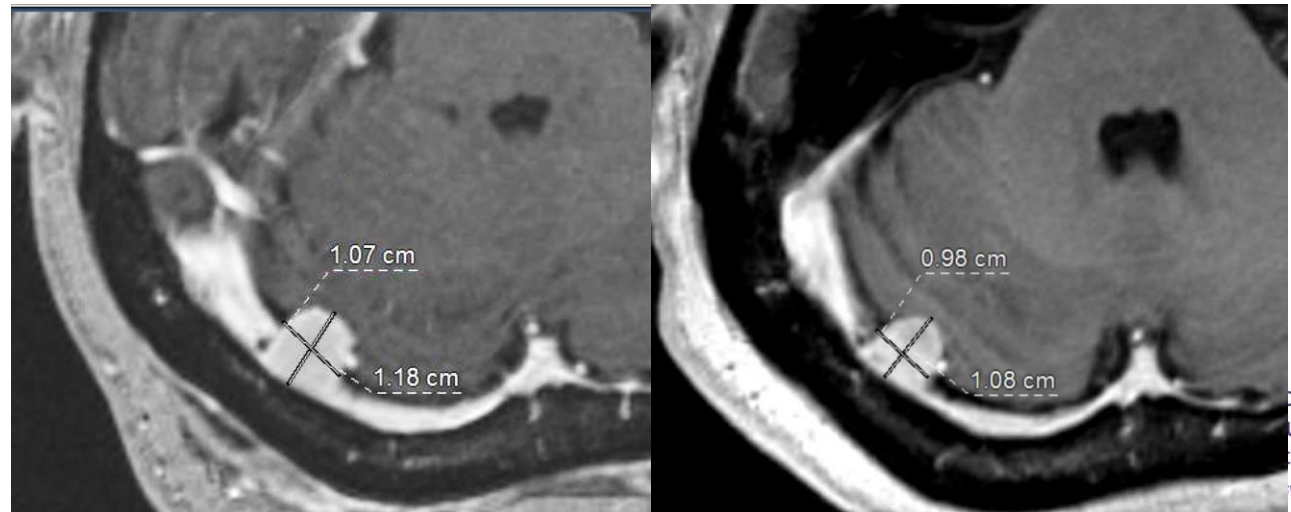
- **90% likelihood of controlling a meningioma with radiation**
- **Rule of Threes**
  - Tumor unchanged 30%
  - Minor shrinkage 30%
  - Major shrinkage 30%
- **May not reverse neurological symptoms (e.g., visual loss)**

# Tumor Response after Radiosurgery

Tumor  
Unchanged



Minor  
Shrinkage



# Side Effects of Radiation



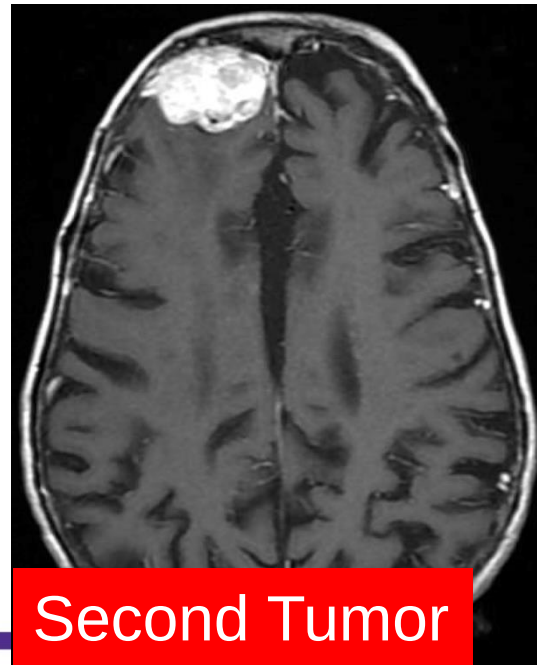
Fatigue



Hair Loss



Scar Tissue



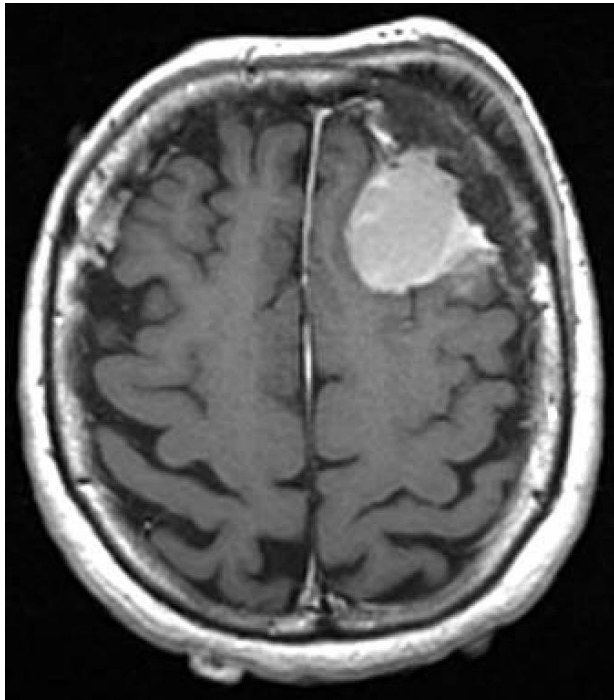
Second Tumor

# Medical Therapy: *Indications*

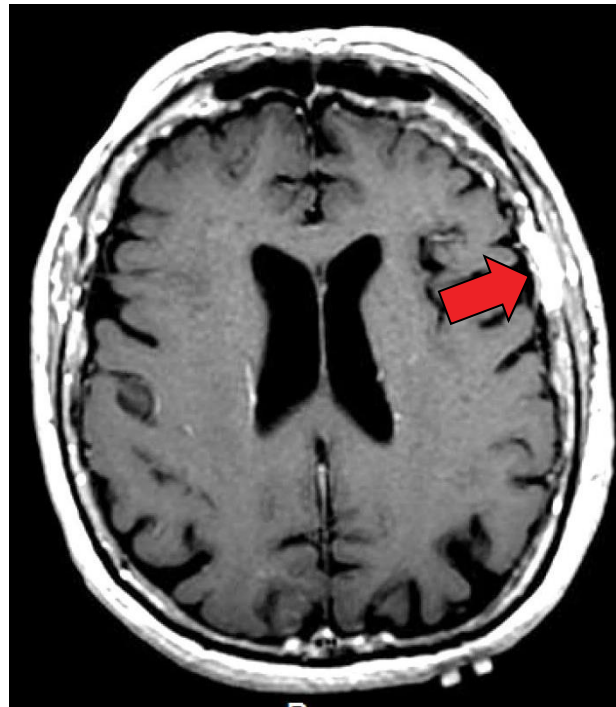
- **Recurrent meningioma after previous surgery and radiation**
- **Multiple meningiomas**
- **Diffuse meningiomas**

# Recurrent Meningioma

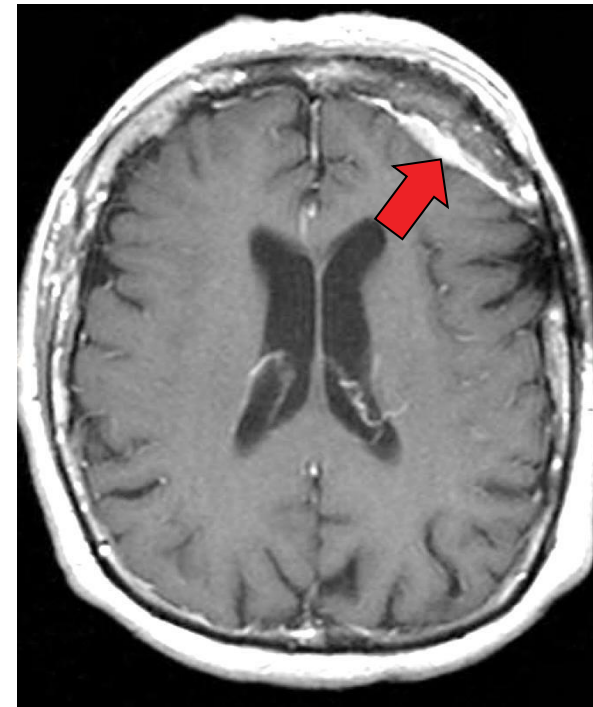
2009



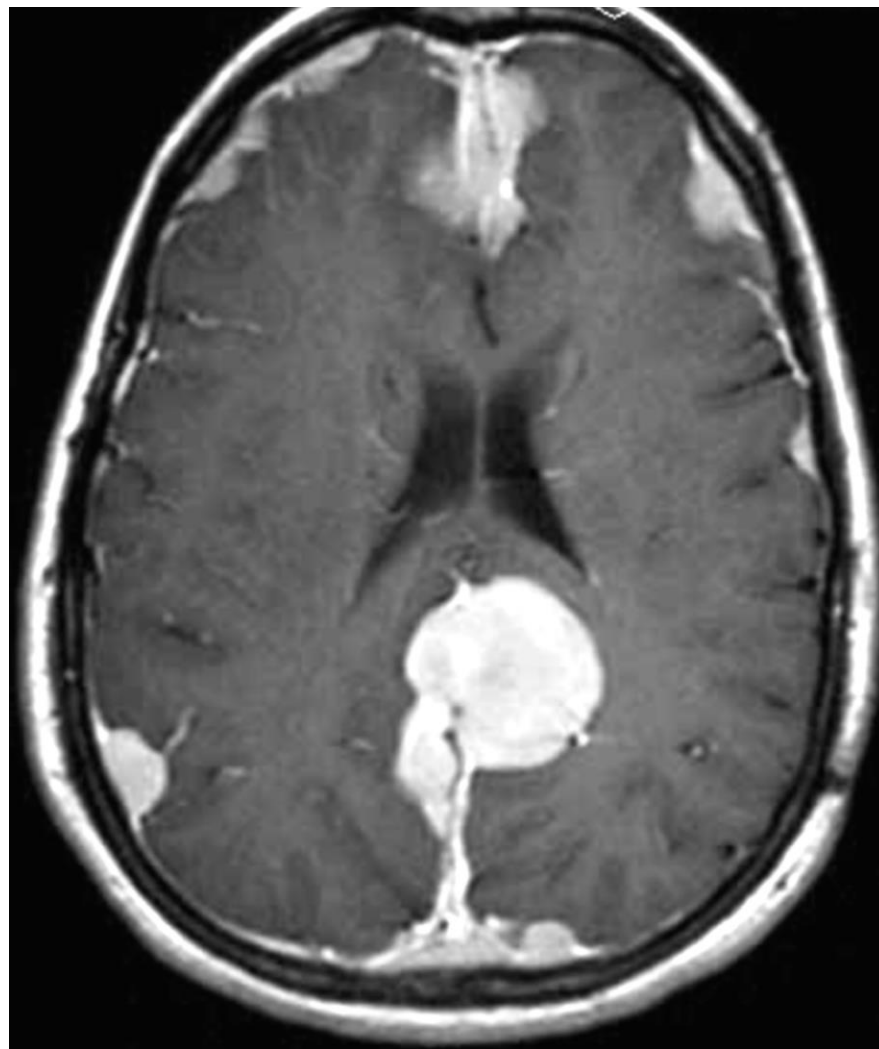
2012



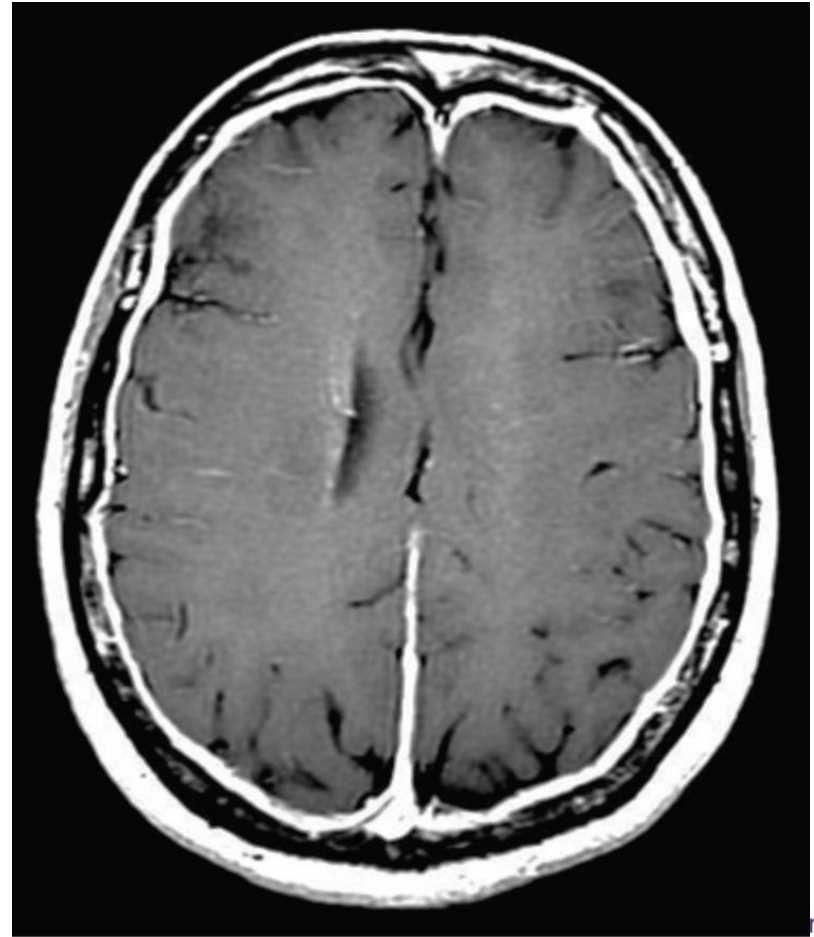
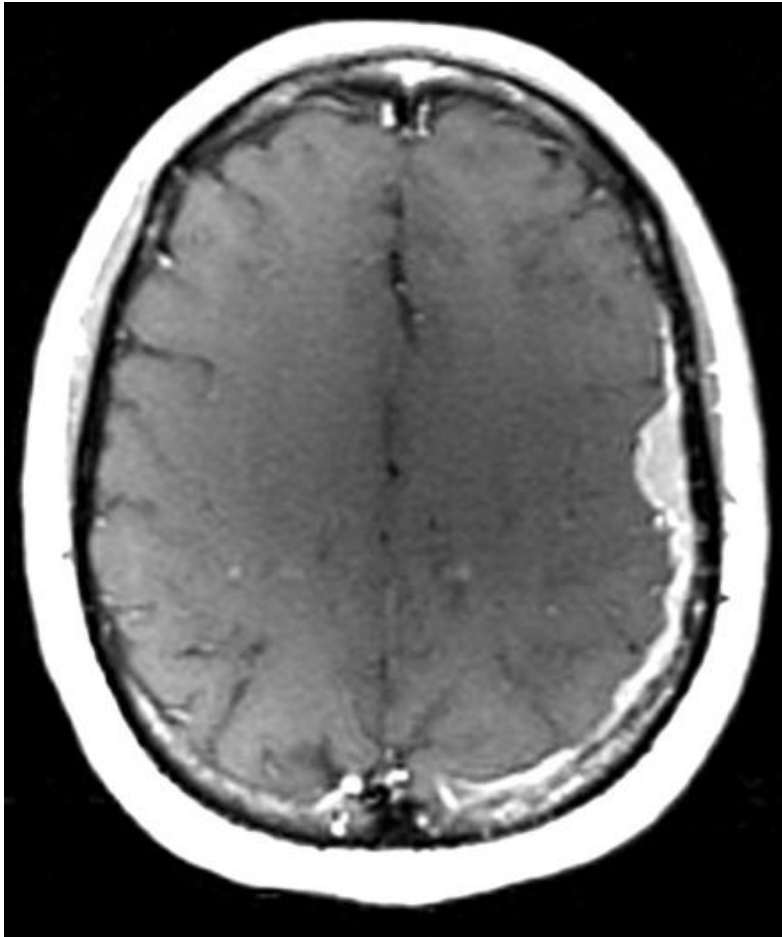
2014



# Multiple Meningiomas



# Diffuse Meningioma



# Approved Medical Therapies

## Hydroxyurea



- Oral
- Promotes cell death pathway

## Alpha Interferon



- Injection
- Inhibits tumor blood vessels

## Sandostatin



- Injection
- Works through hormone receptors

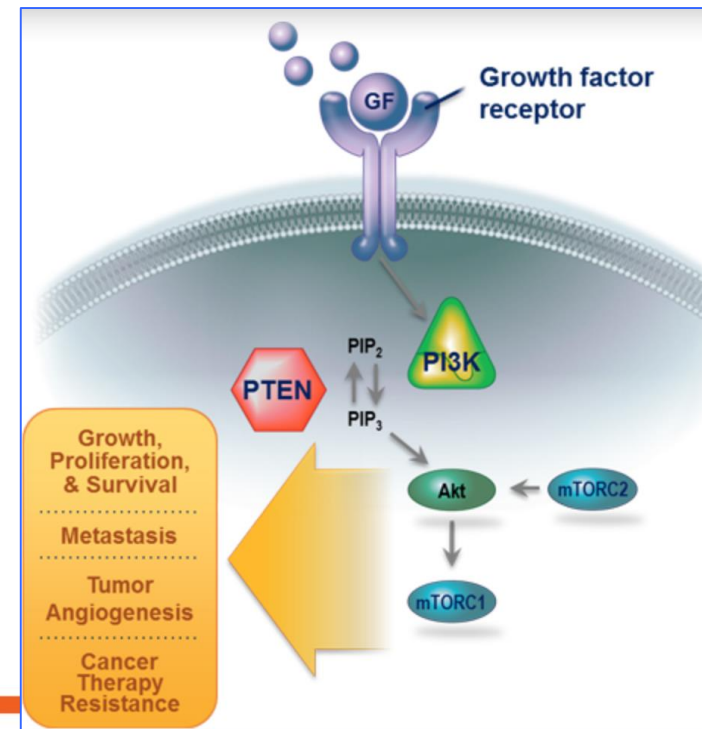
# Future Advances: *Clinical Trials*

## ■ Standard agents

- Hydroxyurea, Alpha Interferon, Sandostatin
- Usually combined with another drug

## ■ Targeted biologic agents

- Everolimus (Afinitor)
- Bevacizumab (Avastin)
- Sunitinib (Sutent)
- And more...



# Future Advances: *NovoTTF*

- Alternating electrical currents disrupt tumor cell division
- Greater benefit for high-grade meningiomas



# Questions?